

2026

Penrith High School
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION



Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11–35 show relevant mathematical reasoning and/or calculations
- Write your NESA ID below, on the Multiple Choice Answer Sheet and the front of Booklets 1, 2, 3 & 4.

Total marks:
100

Section I – 10 marks (pages 3-10)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 11-44)

- Attempt Questions 11–35
- Allow about 2 hours and 45 minutes for this section

NESA NUMBER:

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TEACHER: _____

	Functions	Trigonometric Functions	Calculus	Exponential & Logarithmic Functions	Statistical Analysis	Financial Mathematics
Multiple Choice:	/3 1, 6, 8		/4 2, 4, 9, 10		/3 3, 5, 7	

Total (including Multiple Choice)	Functions	Trigonometric Functions	Calculus	Exponential & Logarithmic Functions	Statistical Analysis	Financial Mathematics	TOTAL
	/11	/22	/18	/11	/22	/18	/102

Section I

10 marks

Attempt questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

- 1 The parabola $y = x^2 + 2$ is translated down 4 units. It is then reflected about the x -axis.

What is the equation of the resulting parabola?

A. $y = -x^2 - 6$

B. $y = -x^2 - 2$

C. $y = -x^2 + 2$

D. $y = -x^2 + 6$

- 2 Calculate $\frac{d}{dx} \left[\int_5^x (g^4 - 2g^3 + g - 1) dg \right]$

A. $x^4 - 2x^3 + x$

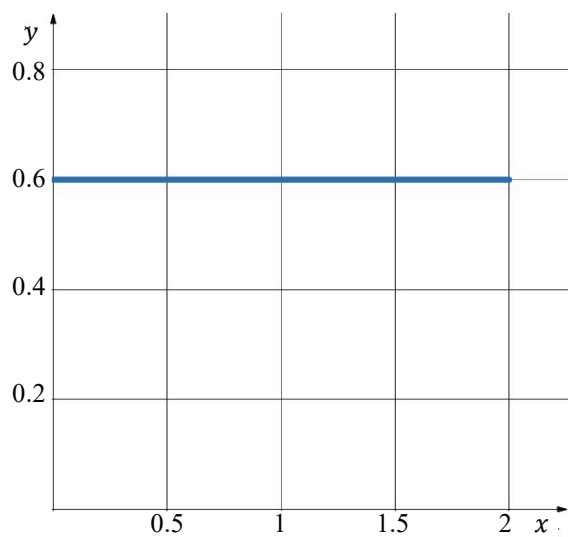
B. $x^4 - 2x^3 + x - 6$

C. $x^4 - 2x^3 + x - 1$

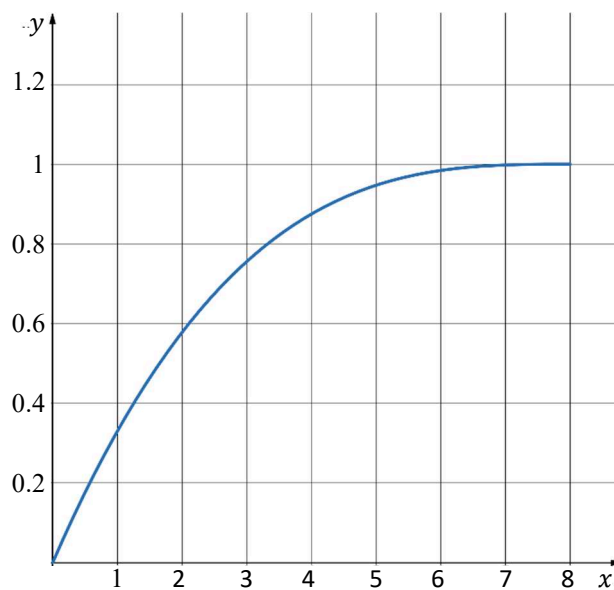
D. $x^4 - 2x^3 + x + C$

- 3 Which of the following could represent a probability density function for a continuous random variable?

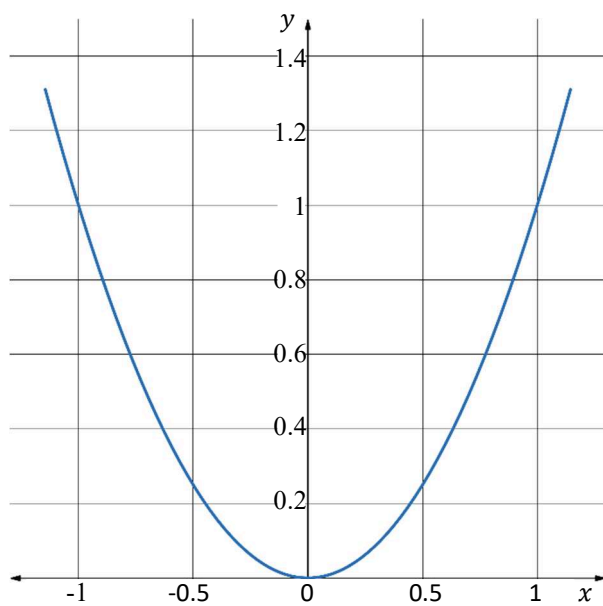
A.



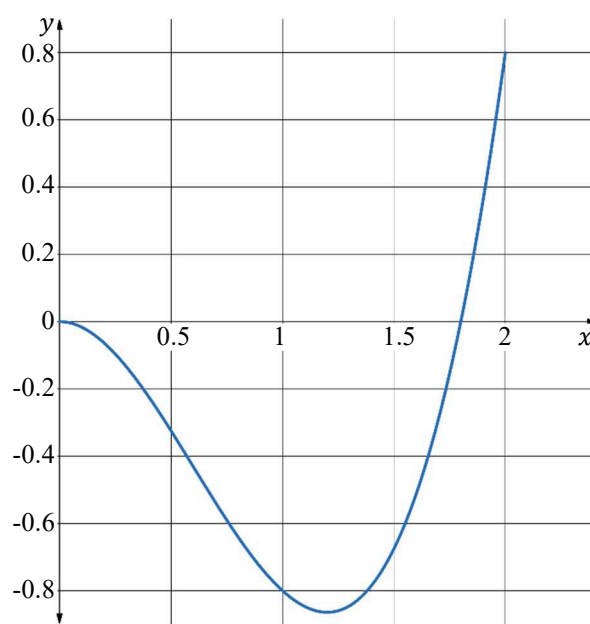
B.



C.



D.



4 Find the angle of inclination of the line $y = -8x - 5$.

- A. 76°
- B. 83°
- C. 97°
- D. 104°

5 A bus company collected data on the number of passengers travelling on a particular route each weekday during the month of June. The table shows the data collected.

<i>Number of passengers</i>	<i>Frequency</i>	<i>Cumulative Frequency</i>
29	2	2
30	5	7
31	6	13
32	8	21
33	1	22

The bus company calculated the mean, median, mode, range and interquartile range for this data.

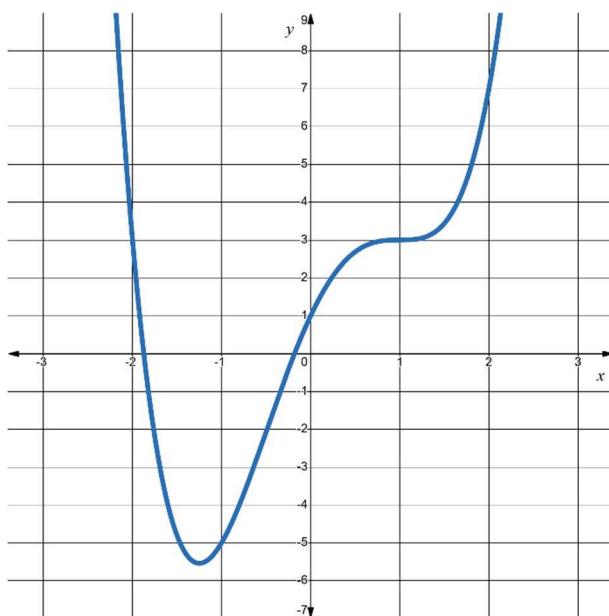
One Friday in June was a public holiday and there was a festival on to celebrate. On this day there were 48 passengers on the bus route. The bus company omitted this data from the table as it was an outlier which would not occur every month.

Which of the statistics that the bus company calculated would this outlier have changed?

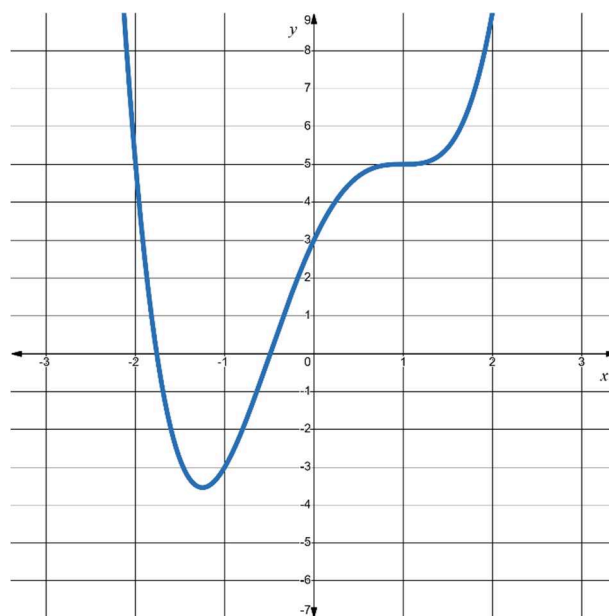
- A. All of the statistics
- B. Mean, median and range only
- C. Mean, mode and range only
- D. Mean and range only

6 Which of the following is the graph of $y = x^4 - x^3 - 3x^2 + 5x + 1$?

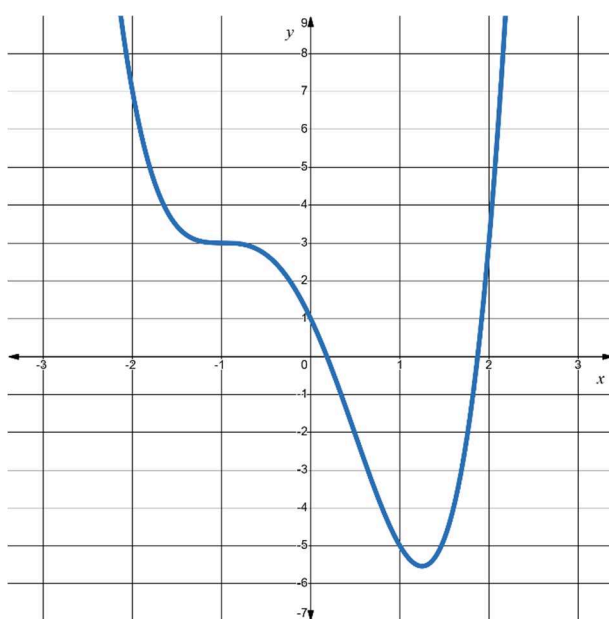
A.



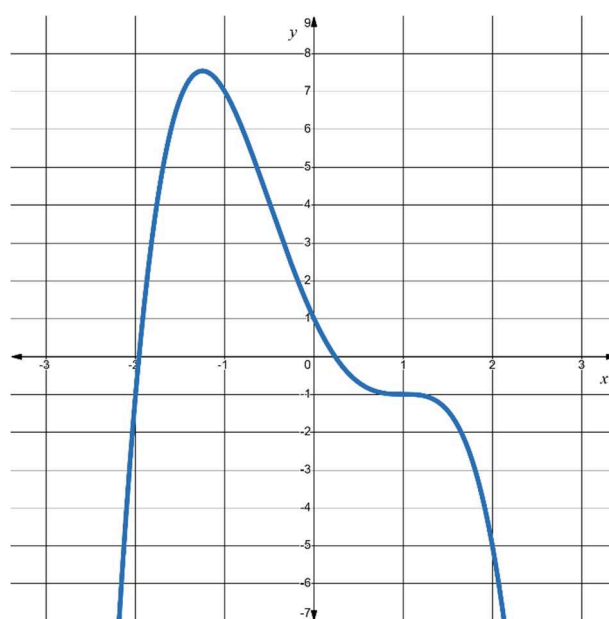
B.



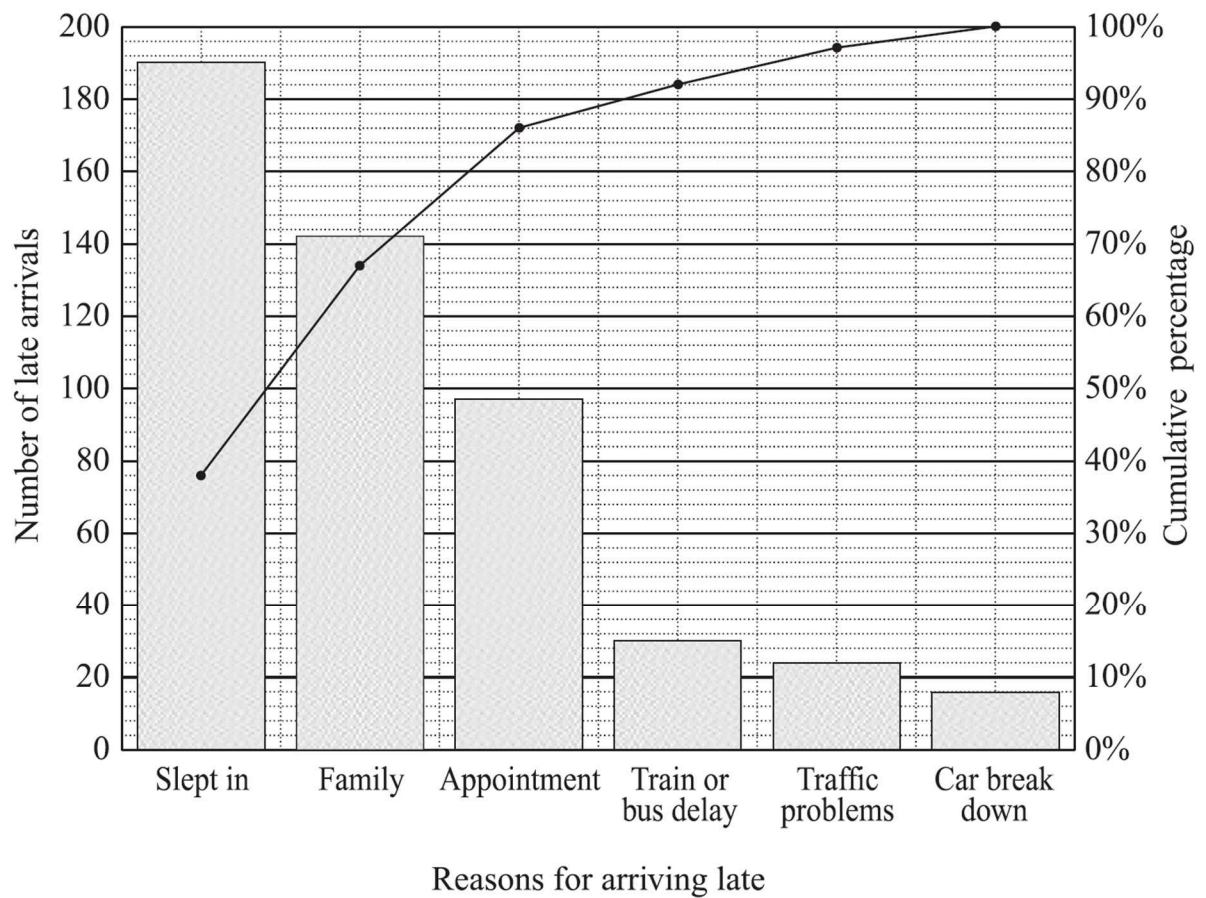
C.



D.



- 7 A school collected data related to the reasons given by students for arriving late. The Pareto chart shows the data collected.



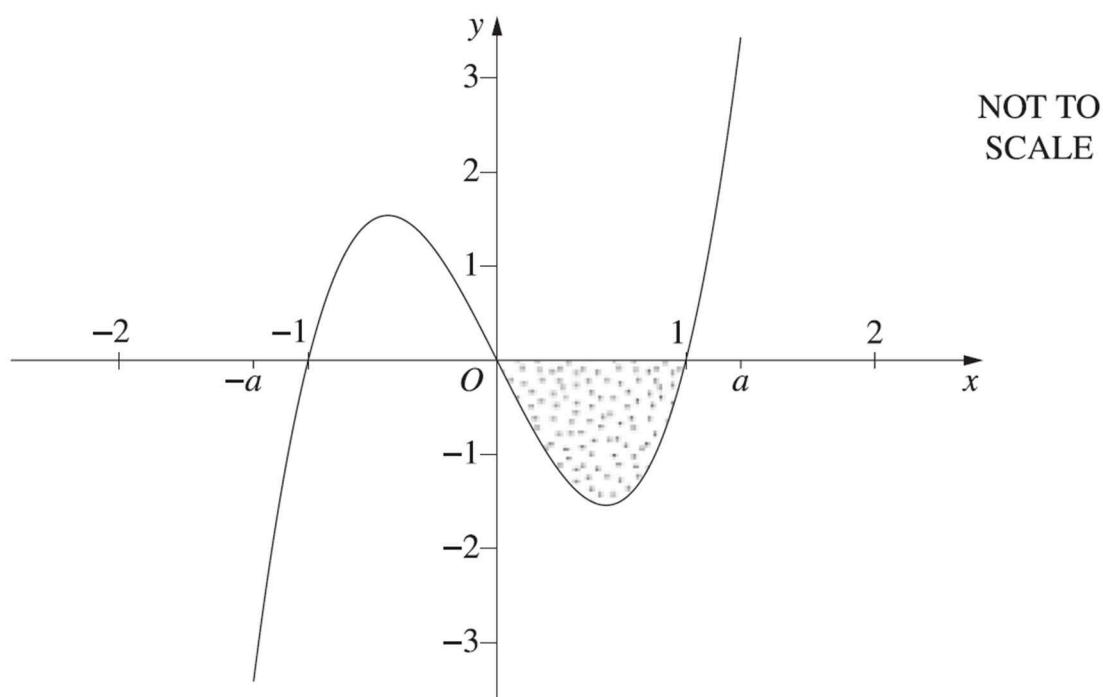
What percentage of students gave the reason 'Traffic problems'?

- A. 2.5%
- B. 5%
- C. 12%
- D. 97%

8 Which of the following describes the graph of $y^2 - 6y + x^2 - 16 = 0$?

- A. A circle with centre at $(0, -3)$ and radius 5 units.
- B. A circle with centre at $(0, 3)$ and radius 5 units.
- C. A circle with centre at $(0, 3)$ and radius 4 units.
- D. A circle with centre at $(3, 0)$ and radius $\sqrt{7}$ units.

9 The diagram shows the graph $y = f(x)$, where $f(x)$ is an odd function. The shaded area is 1 square unit.

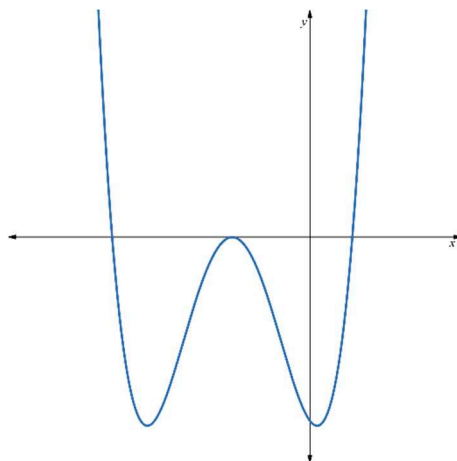


A new function, $g(x)$, is found by taking the graphing of $y = f(x)$ and translating it by 3 units upwards.

Find $\int_{-1}^1 g(x) dx$.

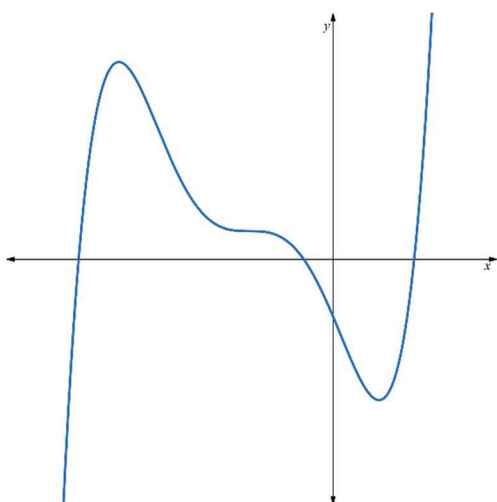
- A. 2
- B. 3
- C. 4
- D. 6

- 10 The graph of a function $y = f'(x)$ is shown below.

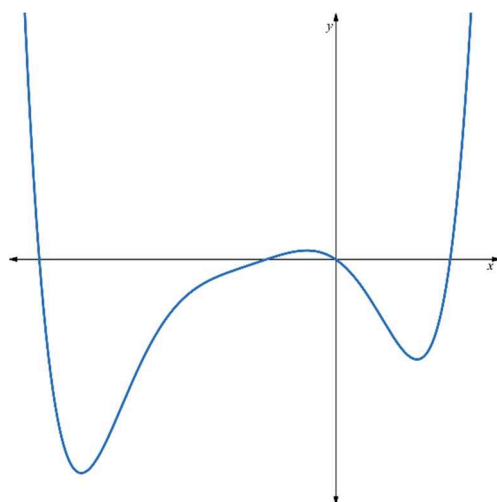


Which of the following is the graph of $y = f(x)$?

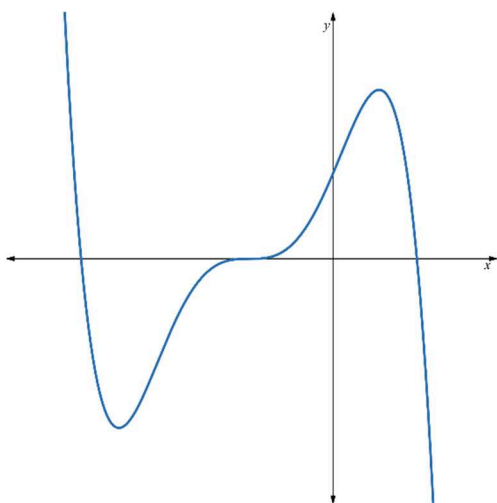
A.



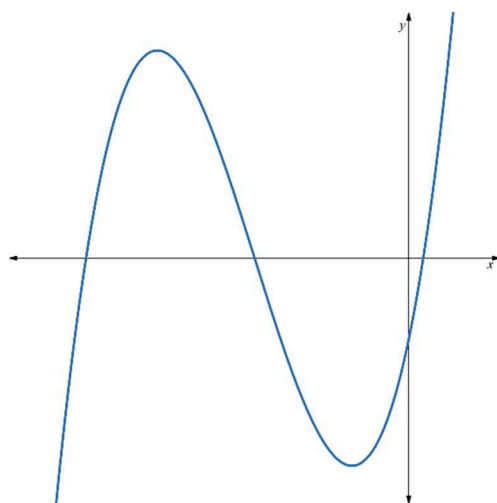
B.



C.



D.



NESA Number:

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Mathematics Advanced Section II Answer Booklet 1

Section II

90 marks

Attempt Questions 11–32

Allow about 2 hours and 45 minutes for this section

Booklet 1 – Attempt Questions 11 – 18 (23 marks)

Booklet 2 – Attempt Questions 19 – 24 (25 marks)

Booklet 3 – Attempt Questions 25 – 30 (22 marks)

Booklet 4 – Attempt Questions 32 – 35 (22 marks)

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
- Write your NESA ID above

Booklet 1	Functions	Trigonometric Functions	Calculus	Exponential & Logarithmic Functions	Statistical Analysis	Financial Mathematics
	16 /4	13, 18 /6	12, 15 /4		11, 14, 17 /9	

Please turn over

Question 11 (2 marks)

A survey of 230 students found that 180 caught a train, 75 caught a bus and 40 caught neither on the way to school.

2

What is the probability that a student caught a bus, given that they also caught a train?

Question 12 (2 marks)

Isaac is training for a marathon. His watch saves his running velocity, $v(t)$, in metres per minute every 20 minutes. The table gives the velocity, $v(t)$ at time t minutes.

2

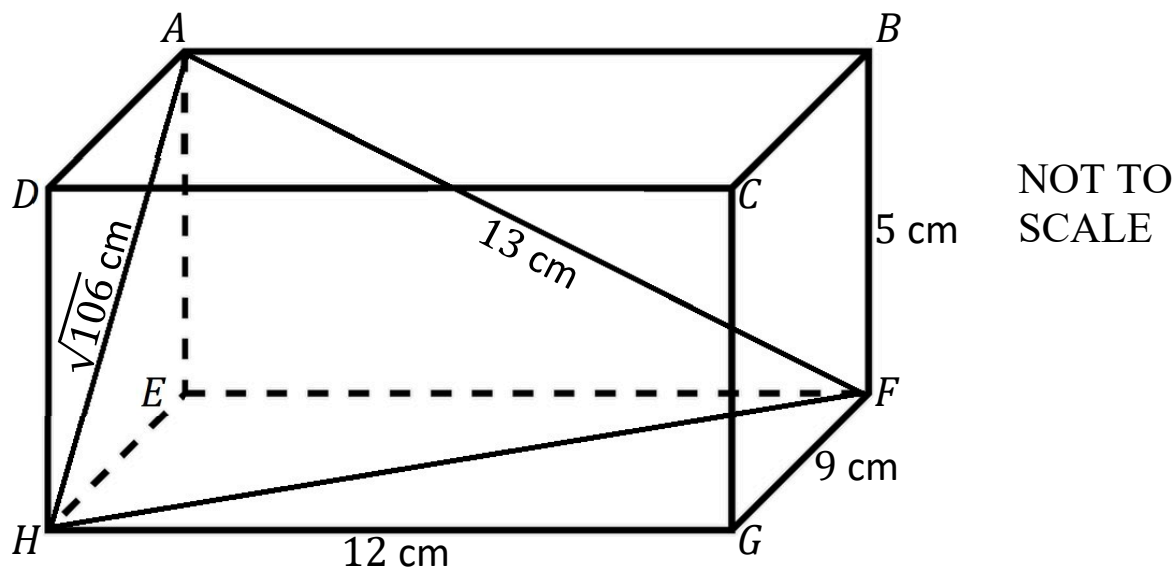
t	0	20	40	60	80
$v(t)$	160	175	178	166	150

Use the trapezoidal rule and the velocity at each of the five time values to find the approximate distance in kilometres that Isaac has run in his 80 minute training session.

Question 13 (3 marks)

The rectangular prism below has $BF = 5$ cm, $FG = 9$ cm, $GH = 12$ cm, $AH = \sqrt{106}$ cm and $AF = 13$ cm, as shown in the diagram below.

3



Find the size of $\angle HAF$, correct to the nearest degree.

Question 14 (4 marks)

- (a) A discrete random variable X has the probability distribution table shown below.

2

$X = x$	3	5	a	12
$P(x)$	0.1	m	0.2	0.4

Find the value of a and m given an expected value of 8.4.

- (b) Hence find the variance of X .

2

Question 15 (2 marks)

Differentiate $f(x) = x^2 - 3x + 2$ using the first principles formula $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

2

Question 16 (4 marks)

For which values of k does $y = kx^2 + (k - 2)x + 1$ have real roots? Provide your answer in simplified exact form.

4

Question 17 (3 marks)

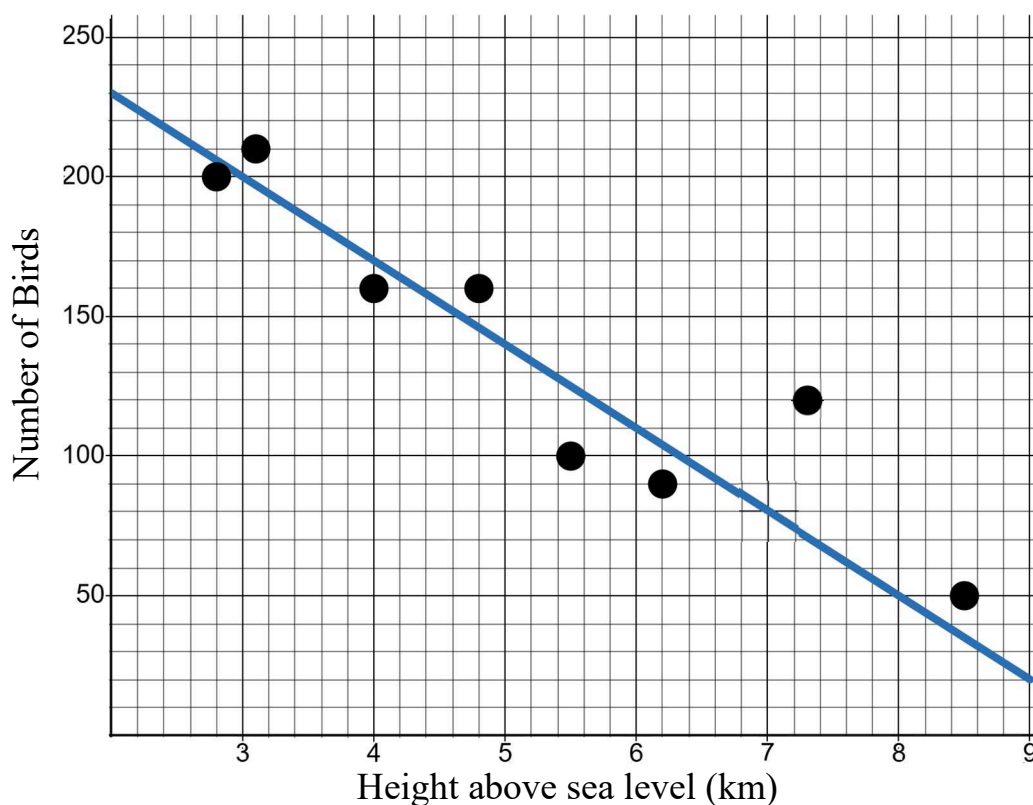
3

Rosanne studies the number of birds B flying above a National Park. She compares the number of birds at different heights h , in kilometres, between 2 and 9 km above sea level. She finds that the data she gathers follows a linear pattern which can be modelled by a least squares regression line with equation

$$B = -30h + c$$

where c is a constant.

Some of the data points that Rosanne collected along with the least squares regression line are shown in the scatterplot below.



Find the value of c , interpret its meaning in the given context and comment on the accuracy of this value as an estimate.

Question 18 (3 marks)

Solve $5 \cos(2x + \frac{\pi}{3}) + 2 = 0$ for $0 \leq x \leq \pi$.

3

[illegible]

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

NESA Number:

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2026

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TRIAL EXAMINATION

Mathematics Advanced

Section II Answer Booklet 2

Booklet 2 – Attempt Questions 19 – 24 (25 marks)

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
- Write your NESA ID above

Booklet 2	Functions	Trigonometric Functions	Calculus	Exponential & Logarithmic Functions	Statistical Analysis	Financial Mathematics
	21 /2		22 /2	23 /7	19, 24 /2	20, 24 /12

Please turn over

Question 19 (2 marks)

Charlie is playing basketball. He has a 13% chance of scoring a goal from beyond the halfway line. How many shots must he take to ensure at least a 97% chance of scoring at least one goal?

2

Question 20 (2 marks)

A table of future value interest factors for an annuity of \$1 is shown.

2

<i>Rate (r)</i> <i>Period (n)</i>	<i>0.005</i>	<i>0.01</i>	<i>0.015</i>	<i>0.02</i>	<i>0.03</i>	<i>0.06</i>
7	7.10588	7.21354	7.32300	7.43428	7.66246	8.39384
28	29.97452	32.12910	34.48148	37.05121	42.93092	68.52811
56	64.44140	74.58098	86.79754	101.55826	141.15377	418.82235
84	104.07393	130.67227	166.17264	213.86661	365.88054	2209.41674

Owais deposits \$2 000 into a savings account every three months for 7 years at an interest rate of 6% per annum, compounding quarterly.

Find the amount of money in Owais' savings account at the end of the 7 years.

Question 21 (2 marks)

At the point $(3, -4)$ on the curve $y = f(x)$, the tangent has equation $y = 2x - 10$. Determine the equation of the tangent to the curve $y = f(-x)$ at the point $(-3, -4)$.

2

Question 22 (2 marks)

Find $\int \frac{3x - 4}{3x^2 - 8x} dx$.

2

Question 23 (7 marks)

- (a) Use a logarithm law to write $\ln \frac{3x-1}{x+2}$ in the form $\ln a - \ln b$, where a and b are algebraic expressions with no fractions.

1

- (b) Hence, show that $y = \ln \frac{3x-1}{x+2}$ has a gradient of 1 at $x = \frac{-5 \pm \sqrt{133}}{6}$.

3

Question 23 continues on the next page.

- (c) (i) Anika thought there was a fault in the answer to Question 23b. She said the answer should be $x = \frac{-5+\sqrt{133}}{6}$ only. However, she is incorrect, there is no error in the answer provided in Question 23b. Anika made an error, as she was solving the question:

1

“Hence, show that $y = \ln a - \ln b$ has a gradient of 1 at $x = \frac{-5 \pm \sqrt{133}}{6}$,”

Instead of the question:

“Hence, show that $y = \ln \frac{3x-1}{x+2}$ has a gradient of 1 at $x = \frac{-5 \pm \sqrt{133}}{6}$,”

Provide the restriction that Anika used that lead to her incorrect answer.

- (c) (ii) Substitute $x = \frac{-5-\sqrt{133}}{6}$ into a and b , provide answers correct to 2 decimal places, and hence explain why $x = \frac{-5-\sqrt{133}}{6}$ is not a solution to Question 23b if you start with the equation $y = \ln a - \ln b$ but it is a solution if you start with the equation $y = \ln \frac{3x-1}{x+2}$.

2

Question 24 (10 marks)

One morning a farmer's crops were attacked by 2 000 locusts (insects). During daylight each day, including the first, the locusts population increases by 15%. Unfortunately, the farmer doesn't realise the severity of the locusts for the first two weeks.

- (a) Show that there are 14 151 locusts at the farm when the farmer realises the severity of the locust plague, to the nearest whole.

1

- (b) The farmer does not wish to spray his farm with pesticide, so he decides to try a locust trap instead. At the start of the 15th day the farmer installs the locust trap, which traps the locust at nighttime. Find the minimum number of locusts per day that the trap must catch to reduce the number of locust from the 15th day onwards.

1

- (c) Once the trap is in use, the number of locusts at the end of the n th day is given by A_n . The locust trap is guaranteed to successfully catch exactly 4 000 locusts per day.

2

- (i) Show that $A_2 = 14\,151(1.15)^2 - 4000(1.15) - 4000$.

Question 24 continues on the next page.

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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NESA Number:

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TRIAL EXAMINATION

Mathematics Advanced

Section II Answer Booklet 3

Booklet 3 – Attempt Questions 25 – 30 (22 marks)

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
- Write your NESA ID above

Booklet 3	Functions	Trigonometric Functions	Calculus	Exponential & Logarithmic Functions	Statistical Analysis	Financial Mathematics
	27 /2	25, 30 /6	28 /4		29 /8	26 /2

Please turn over

Question 25 (2 marks)

Find x , given that $\sin(x + 10)^\circ = \cos(3x + 20)^\circ$ and $0^\circ \leq x \leq 90^\circ$.

2

Question 26 (2 marks)

A computer program is designed to select any positive integer x . The probability distribution shown in the table below, where m is a constant, shows the probabilities of selecting the first four positive integers only.

2

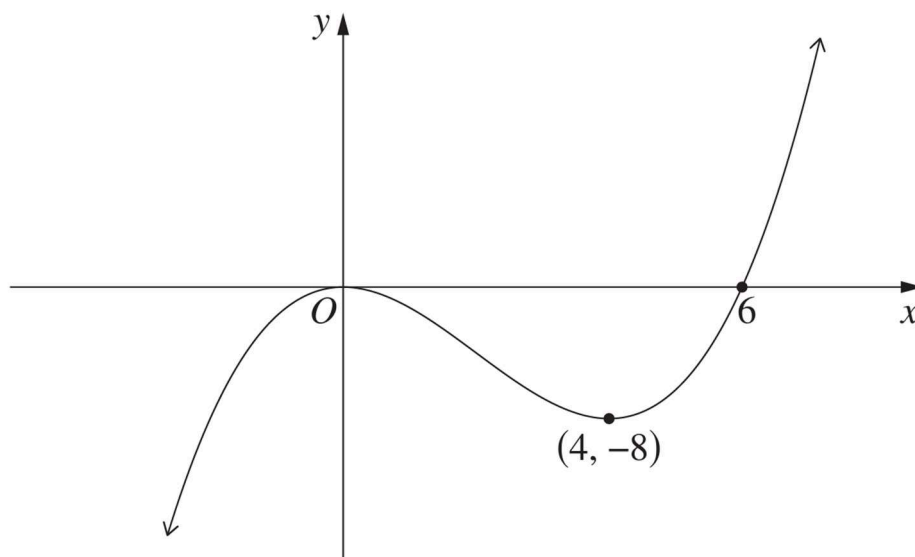
x	1	2	3	4	...
$P(x)$	$3m$	$6m^2$	$12m^3$	$24m^4$	...

Find the value of m .

Question 27 (2 marks)

Consider the graph of $y = f(x)$ as shown.

2



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SCALE

Sketch the graph of $y = f(2x + 4)$, labelling the x -intercepts and the coordinates of the turning points.

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4

4

where t is the time in minutes that have been played in the match and $0 \leq t \leq 90$.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page, typical of notebook or legal stationery. There are no margins, text, or other markings on the paper.

Question 29 (8 marks)

Elena, the owner of Elena's Escape Rooms, records the time taken in minutes for each group to complete her escape room. She recognises this data as a continuous random variable, X , which can be modelled by the probability density function

$$f(x) = \begin{cases} -\frac{1}{4500}(x-10)(x-40) & 10 \leq x \leq 40 \\ 0 & \text{elsewhere} \end{cases}$$

- (a) Find the mode time taken to complete the escape room at Elena's Escape Rooms.

2

- (b) (i) Show that the cumulative density function for the time taken to complete the escape room at Elena's Escape Rooms is given by

2

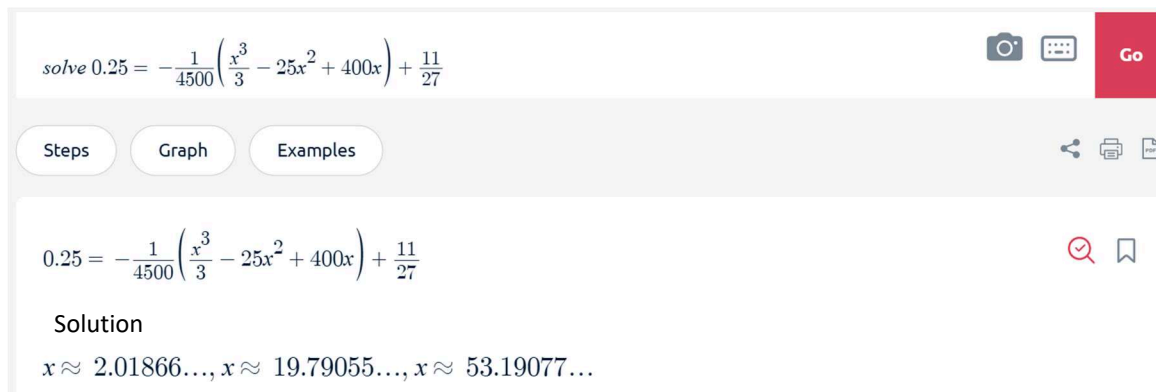
$$F(x) = -\frac{1}{4500}\left(\frac{x^3}{3} - 25x^2 + 400x\right) + \frac{11}{27}$$

Questions 29 continues on the next page.

- (b) (ii) Elena used the cumulative density function found above and an online calculator to find the 25th and 75th percentiles for her escape room. Two screenshots of the calculations made by the online calculator are shown below.

1

Screenshot 1:



$$\text{solve } 0.25 = -\frac{1}{4500} \left(\frac{x^3}{3} - 25x^2 + 400x \right) + \frac{11}{27}$$

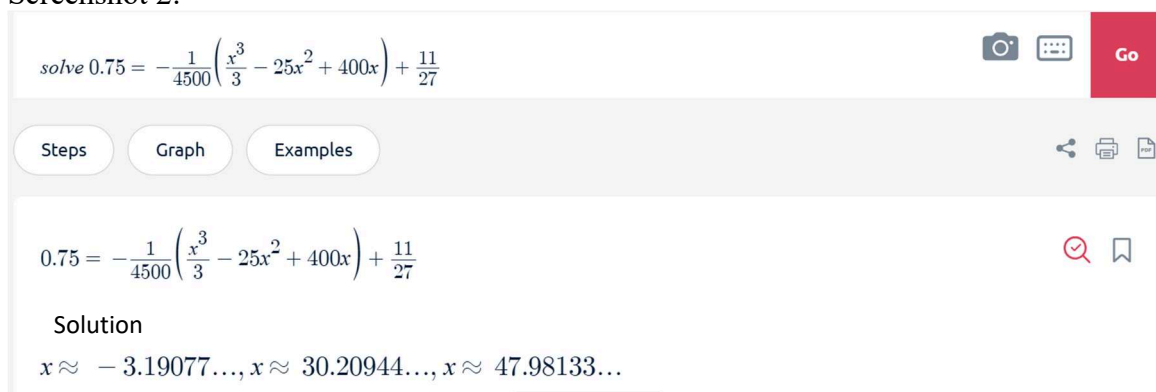
Steps Graph Examples

$$0.25 = -\frac{1}{4500} \left(\frac{x^3}{3} - 25x^2 + 400x \right) + \frac{11}{27}$$

Solution

$x \approx 2.01866..., x \approx 19.79055..., x \approx 53.19077...$

Screenshot 2:



$$\text{solve } 0.75 = -\frac{1}{4500} \left(\frac{x^3}{3} - 25x^2 + 400x \right) + \frac{11}{27}$$

Steps Graph Examples

$$0.75 = -\frac{1}{4500} \left(\frac{x^3}{3} - 25x^2 + 400x \right) + \frac{11}{27}$$

Solution

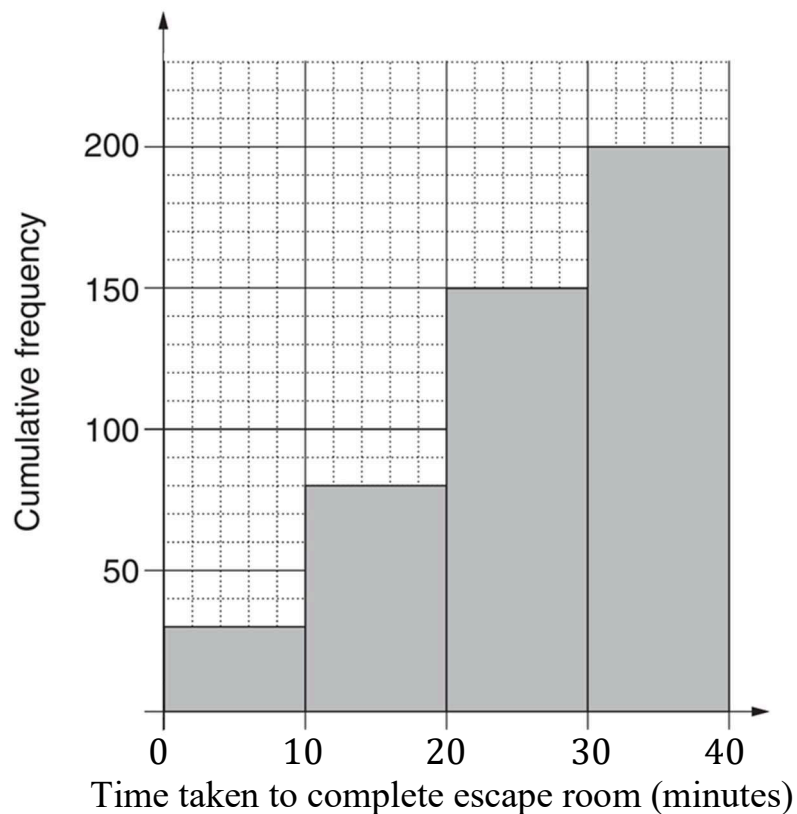
$x \approx -3.19077..., x \approx 30.20944..., x \approx 47.98133...$

Using the screenshots above, calculate the interquartile range for the time taken to complete Elena's Escape Room.

Questions 29 continues on the next page.

- (c) (i) A different escape room company, Escape From Axel, also gathers data on the time its customers take to complete their escape room. Axel displays the data in the cumulative frequency histogram below.

2



Calculate the interquartile range for the time taken to complete the escape room at Escape From Axel.

- (c) (ii) Find the modal time group for the time taken to complete the escape room at Escape From Axel.

1

Question 30 (4 marks)

Identify the period, amplitude, phase shift and vertical shift of the graph formed by the equation $y = 12\cos(\frac{\pi}{3}(x - 2)) + 4$.

4

[illegible]

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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NESA Number:

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Mathematics Advanced

Section II Answer Booklet 4

Booklet 4 – Attempt Questions 31 – 35 (22 marks)

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
- Write your NESA ID above

Booklet 4	Functions	Trigonometric Functions	Calculus	Exponential & Logarithmic Functions	Statistical Analysis	Financial Mathematics
		32, 35 /10	33 /4	31 /4		34 /4

Please turn over

Question 31 (4 marks)

Given $f(x) = 16^x$ and $g(x) = \frac{1}{2}x + \frac{3}{4}$ show that the derivative of $y = f(g(x))$ is $\ln 2 (2^{2x+4})$.

4

[illegible]

Question 32 (3 marks)

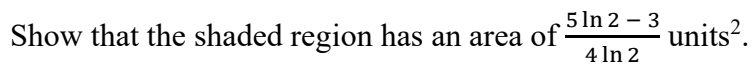
Find $\int \frac{\tan^4 x}{1 - \sec^2 x} dx$.

3

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

4

4

[illegible]

4

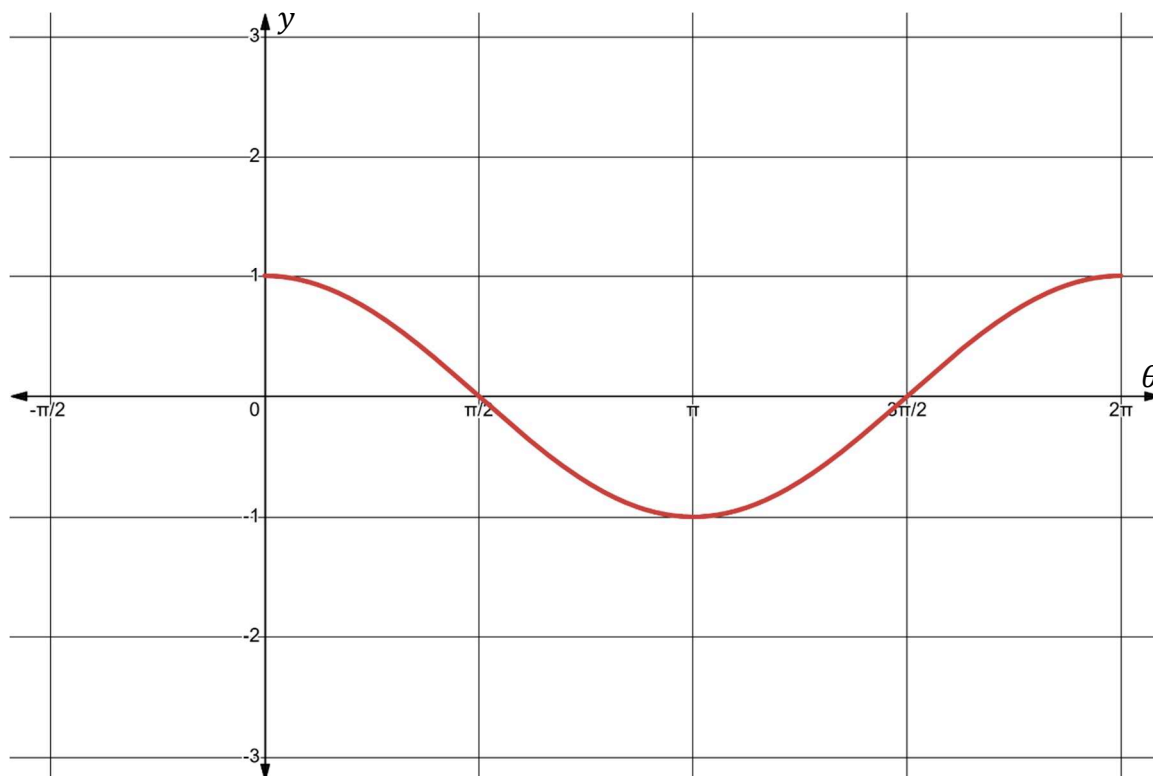
The diagram shows a sector of a circle with a central angle θ . The sector is divided into seven concentric annular regions, labeled 'Landing Zone 1' through 'Landing Zone n' from the innermost to the outermost. The innermost zone (Zone 1) has a width of 8 m. The radius of the innermost zone is 20 m. The outermost zone is labeled 'Landing Zone n'.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 35 (7 marks)

- (a) The graph of $y = \cos \theta$ for $0 \leq \theta \leq 2\pi$ is shown below. On the same Cartesian Plane, sketch the graph of $y = -\sin \theta + 1$ for $0 < \theta < 2\pi$.

3



- (b) Hence or otherwise, find any point(s) of intersection of $y = \cos \theta$ and $y = -\sin \theta + 1$ for $0 < \theta < 2\pi$.

2

Question 35 continues on the next page

2

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Section II extra writing space

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Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11–35 show relevant mathematical reasoning and/or calculations
- Write your NESA ID below, on the Multiple Choice Answer Sheet and the front of Booklets 1, 2, 3 & 4.

Total marks:
100

Section I – 10 marks (pages 3-10)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 11-44)

- Attempt Questions 11–35
- Allow about 2 hours and 45 minutes for this section

NESA NUMBER:

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TEACHER:

Solutions

	Functions	Trigonometric Functions	Calculus	Exponential & Logarithmic Functions	Statistical Analysis	Financial Mathematics
Multiple Choice:	/3 1, 6, 8		/4 2, 4, 9, 10		/3 3, 5, 7	

Total (including Multiple Choice)	Functions	Trigonometric Functions	Calculus	Exponential & Logarithmic Functions	Statistical Analysis	Financial Mathematics	TOTAL
	/11	/22	/18	/9	/22	/18	/100

Section I

10 marks

Attempt questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

- 1 The parabola $y = x^2 + 2$ is translated down 4 units. It is then reflected about the x -axis.

What is the equation of the resulting parabola?

A. $y = -x^2 - 6$

B. $y = -x^2 - 2$

☒ C. $y = -x^2 + 2$

D. $y = -x^2 + 6$

$$y = (x^2 + 2) - 4$$
$$= x^2 - 2$$

$$y = -(x^2 - 2)$$
$$= -x^2 + 2$$

- 2 Calculate $\frac{d}{dx} \left[\int_5^x (g^4 - 2g^3 + g - 1) dg \right]$

A. $x^4 - 2x^3 + x$

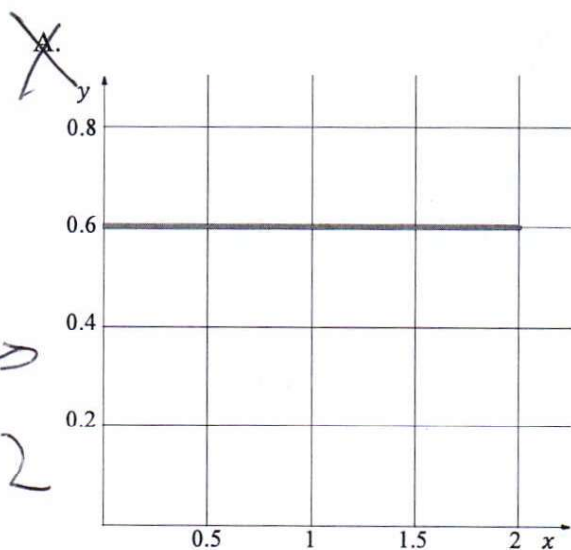
B. $x^4 - 2x^3 + x - 6$

☒ C. $x^4 - 2x^3 + x - 1$

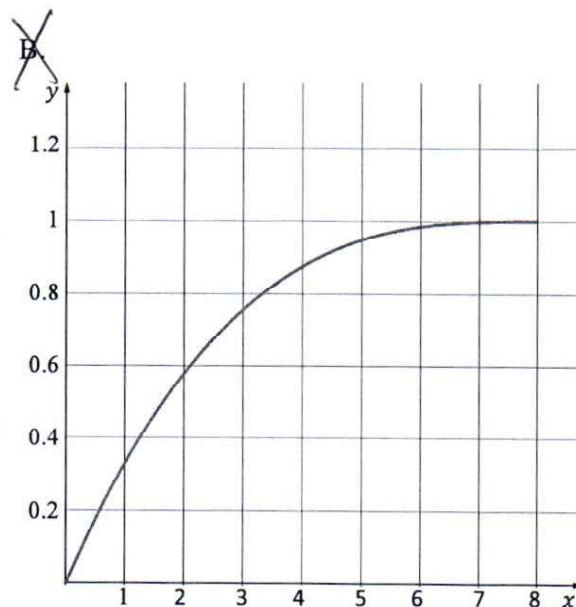
D. $x^4 - 2x^3 + x + C$

By the fundamental
theorem of calculus,

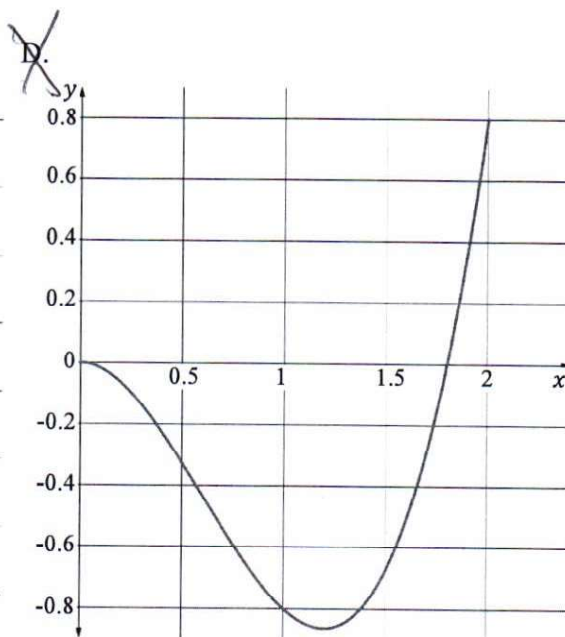
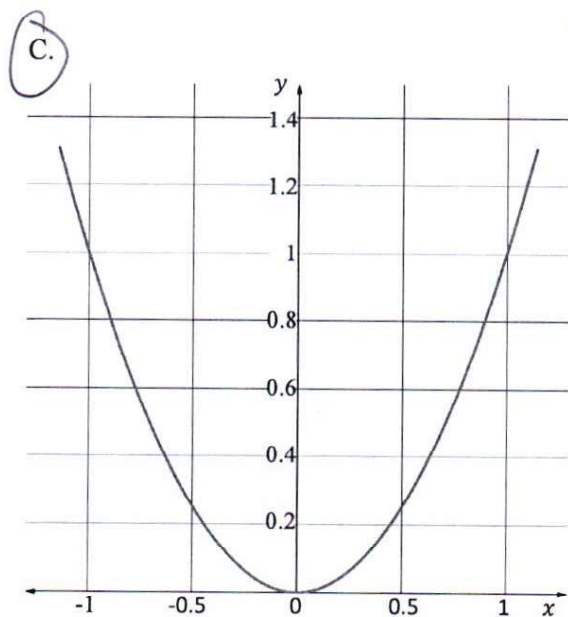
- 3 Which of the following could represent a probability density function for a continuous random variable?



Area $\rightarrow$
 $= 0.6 \times 2$
 $= 1.2$
 $\neq 1$



Area
 is clearly
 greater
 than 1



$\uparrow$

By process of elimination
 this is the answer.
 Note x can be negative
 in a PDF.

$\uparrow$

$y \geq 0$ in a PDF

- 4 Find the angle of inclination of the line $y = -8x - 5$.

A. 76°

B. 83°

☒ C. 97°

D. 104°

$$\begin{aligned} m &= \tan \theta \\ -8 &= \tan \theta \\ \tan^{-1}(-8) &= \theta \\ \theta &= -83 \\ 180 - 83 &= 97 \end{aligned}$$

- 5 A bus company collected data on the number of passengers travelling on a particular route each weekday during the month of June. The table shows the data collected.

Number of passengers	Frequency	Cumulative Frequency
29	2	2
30	5	7
31	6	13
32	8	21
33	1	22

The bus company calculated the mean, median, mode, range and interquartile range for this data.

One Friday in June was a public holiday and there was a festival on to celebrate. On this day there were 48 passengers on the bus route. The bus company omitted this data from the table as it was an outlier which would not occur every month.

Which of the statistics that the bus company calculated would this outlier have changed?

- ☒ A. All of the statistics
- ☒ B. Mean, median and range only
- ☒ C. Mean, mode and range only
- ☒ D. Mean and range only

An outlier will not change mode.
So eliminate A and C.

Now B and C differ only by median. So will it change the median?

At the moment there are 22 scores, so median is the average of 11th and 12th scores (31)

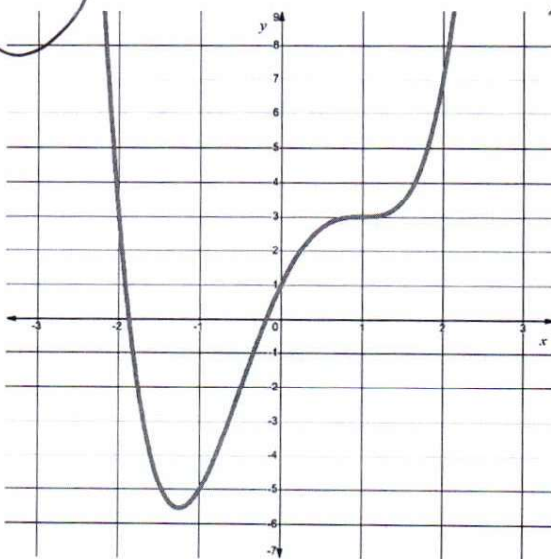
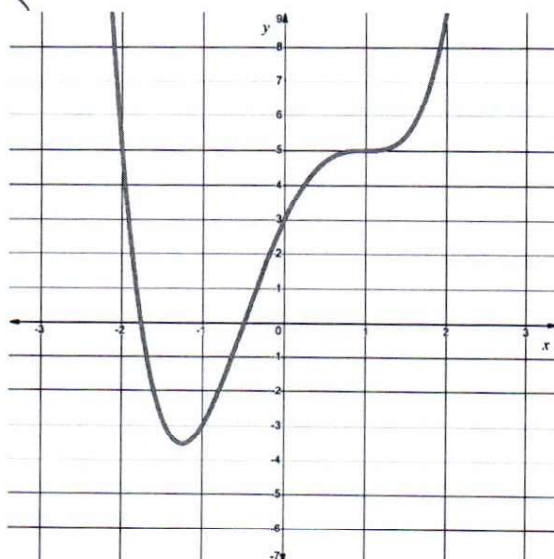
If another score was added there will be 23 scores, so the median will be the 12th score. Which is also 31.
So median won't change.

6

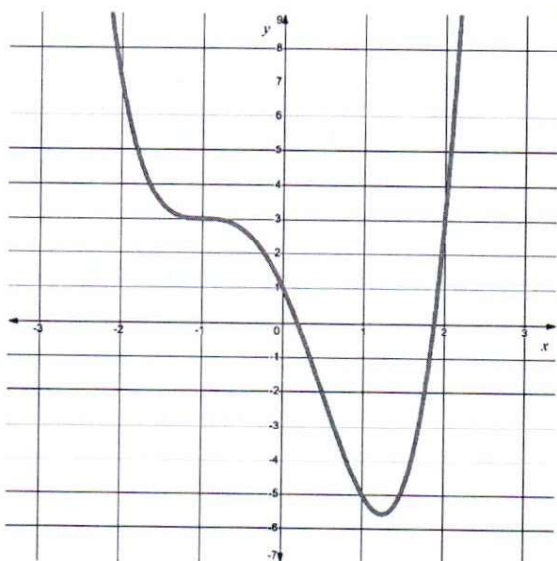
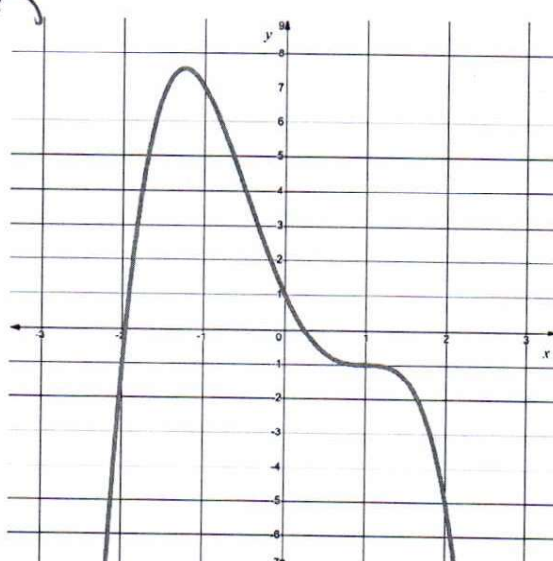
Which of the following is the graph of $y = x^4 - x^3 - 3x^2 + 5x + 1$?

y-intercept
should be 1
so eliminate B.

A.

~~B.~~

C.

~~D.~~

To choose between A or C
We can simply substitute
an x -value in.

Eg. $x = 1$

$$y = 1^4 - 1^3 - 3(1)^2 + 5(1) + 1$$

$$= 3$$

so A

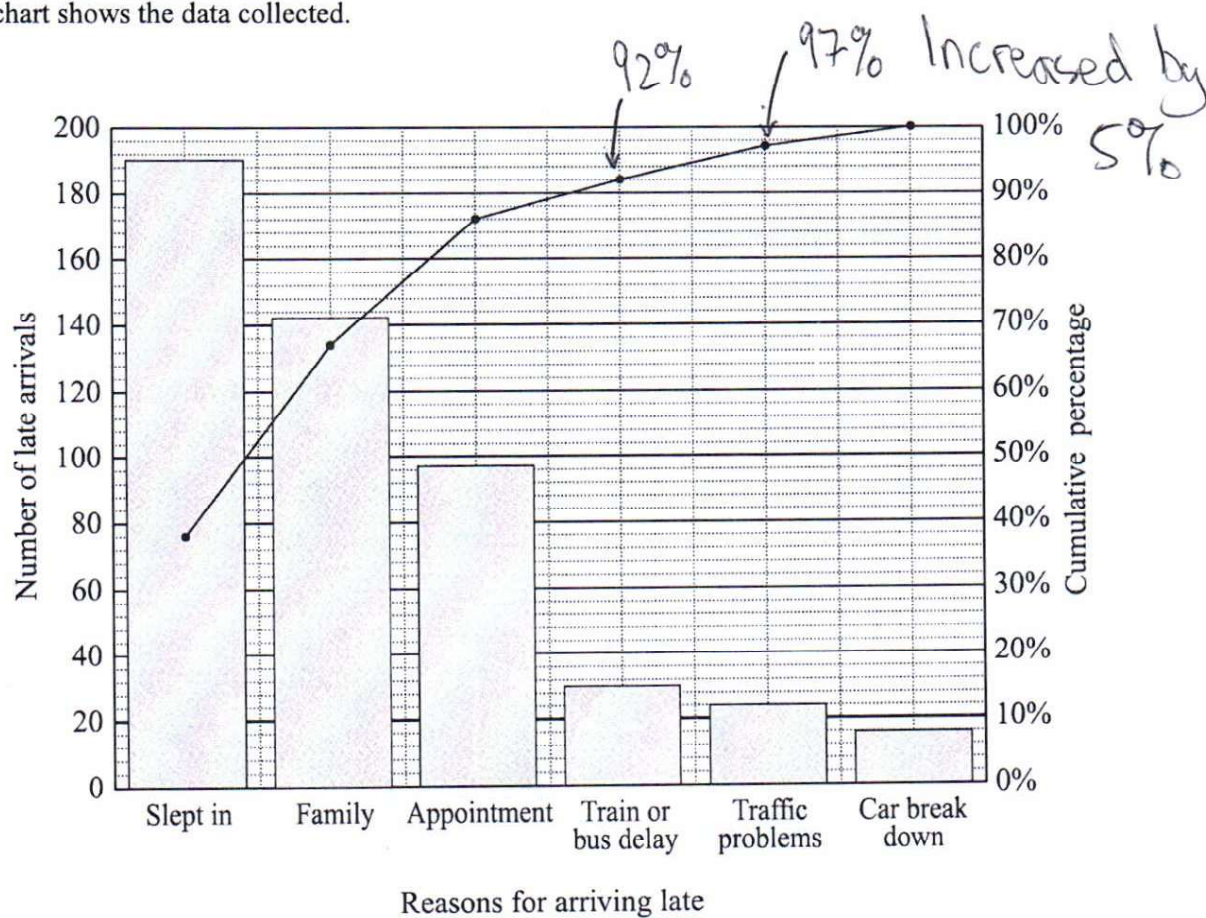
$$a = 1$$

$$> 0$$

so edges should
be up

so eliminate D.

- 7 A school collected data related to the reasons given by students for arriving late. The Pareto chart shows the data collected.



What percentage of students gave the reason 'Traffic problems'?

- A. 2.5%
- ☒ B. 5%
- C. 12%
- D. 97%

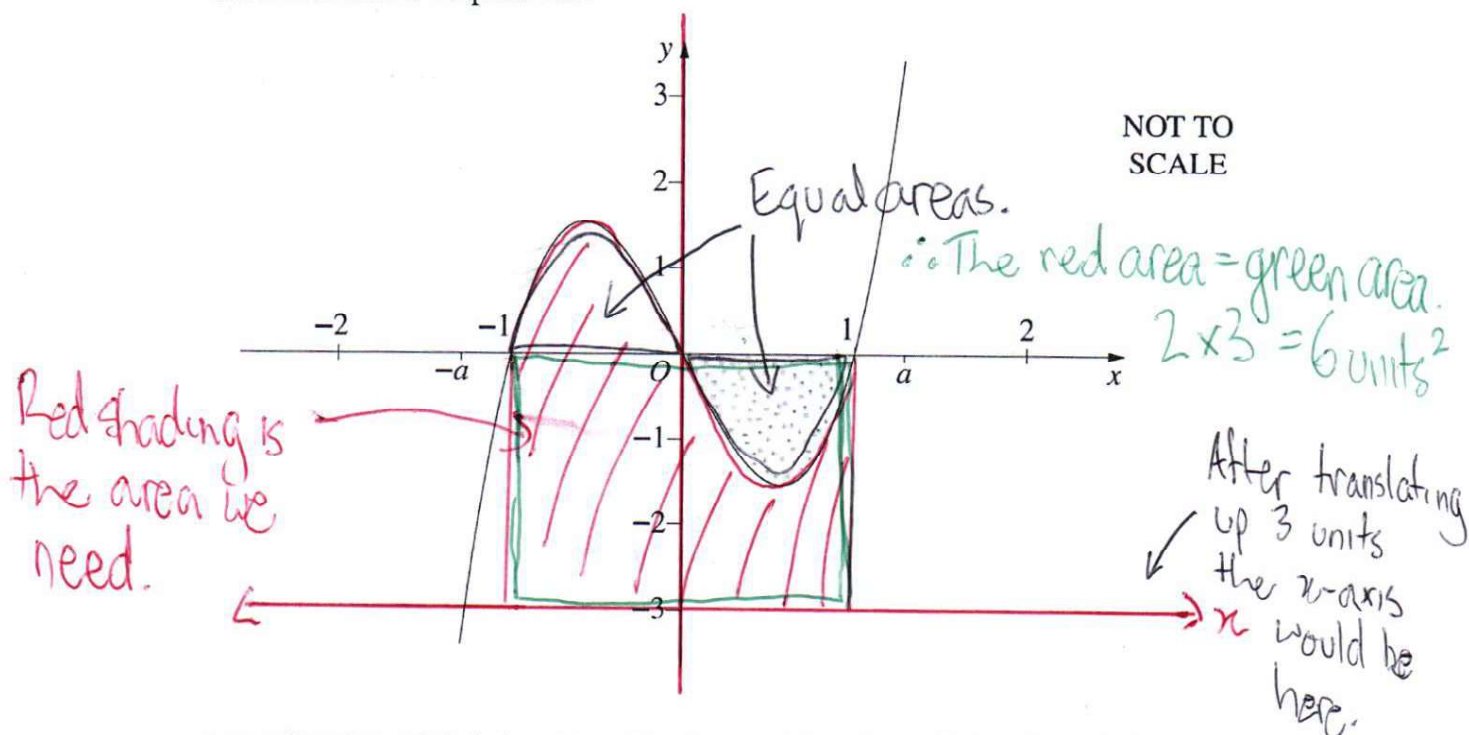
8 Which of the following describes the graph of $y^2 - 6y + x^2 - 16 = 0$?

- A. A circle with centre at $(0, -3)$ and radius 5 units.
- ☒ B. A circle with centre at $(0, 3)$ and radius 5 units.
- C. A circle with centre at $(0, 3)$ and radius 4 units.
- D. A circle with centre at $(3, 0)$ and radius $\sqrt{7}$ units.

$$y^2 - 6y + \left(\frac{6}{2}\right)^2 + x^2 = 16 + \left(\frac{6}{2}\right)^2$$

$$(y - 3)^2 + x^2 = 25$$

9 The diagram shows the graph $y = f(x)$, where $f(x)$ is an odd function. The shaded area is 1 square unit.

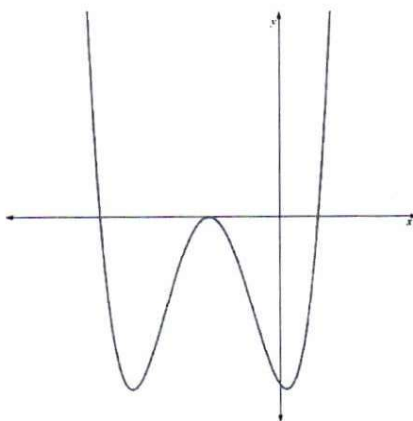


A new function, $g(x)$, is found by taking the graphing of $y = f(x)$ and translating it by 3 units upwards.

Find $\int_{-1}^1 g(x) dx$.

- A. 2
- B. 3
- C. 4
- ☒ D. 6

- 10 The graph of a function $y = f'(x)$ is shown below.



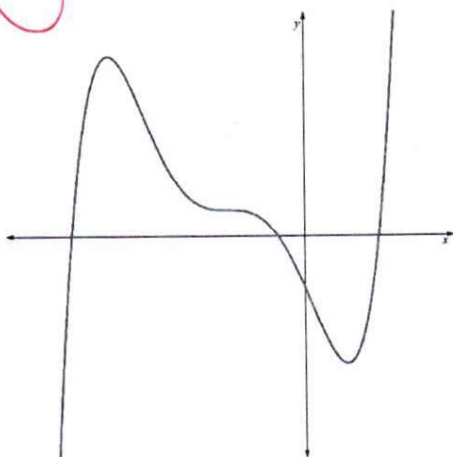
*y-intercepts turn into stationary points.
So eliminate B and D*

On the edges the y values of $y = f'(x)$ are positive, so the gradients of $y = f(x)$ should be positive (uphill) on the edges.

∴ A.

Which of the following is the graph of $y = f(x)$?

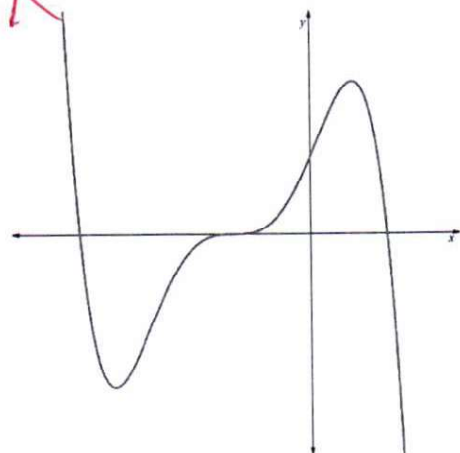
A.



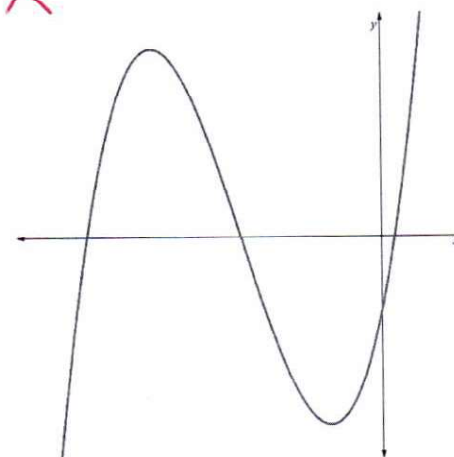
~~B.~~



~~C.~~



~~D.~~



NESA Number:

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2026

**Penrith High School
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Mathematics Advanced Section II Answer Booklet 1

Section II

90 marks

Attempt Questions 11–32

Allow about 2 hours and 45 minutes for this section

Booklet 1 – Attempt Questions 11 – 18 (23 marks)

Booklet 2 – Attempt Questions 19 – 24 (23 marks)

Booklet 3 – Attempt Questions 25 – 30 (22 marks)

Booklet 4 – Attempt Questions 32 – 35 (22 marks)

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
- Write your NESA ID above

Booklet 1	Functions	Trigonometric Functions	Calculus	Exponential & Logarithmic Functions	Statistical Analysis	Financial Mathematics
	16 /4	13, 18 /6	12, 15 /4		11, 14, 17 /9	

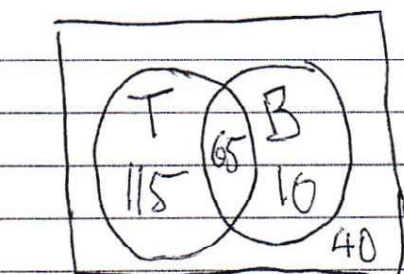
Please turn over

Question 11 (2 marks)

A survey of 230 students found that 180 caught a train, 75 caught a bus and 40 caught neither on the way to school.

2

What is the probability that a student caught a bus, given that they also caught a train?



$$230 - 40 = 190 \text{ left for the circles.}$$

$$180 + 75 = 255$$

$$255 - 190 = 65 \text{ double counted}$$

$$\therefore 65 \text{ in intersection.}$$

$$P(B|T) = \frac{65}{180}$$

comments: Many students did the question correctly.

- Few mistakes include incorrect numbers in the Venn diagram and hence, in working
- did not include 40 in any calculations.
- Overall, okay

Question 12 (2 marks)

Isaac is training for a marathon. His watch saves his running velocity, $v(t)$, in metres per minute every 20 minutes. The table gives the velocity, $v(t)$ at time t minutes.

2

t	0	20	40	60	80
$v(t)$	160	175	178	166	150

Use the trapezoidal rule and the velocity at each of the five time values to find the approximate distance in kilometres that Isaac has run in his 80 minute training session.

$$\frac{b-a}{2n} \{ f(a) + f(b) + 2[f(x_1) + f(x_2) + f(x_3)] \}$$

$$\frac{80-0}{2 \times 4} \{ 160 + 150 + 2[175 + 178 + 166] \}$$

$$= 13480 \text{ m} = 13.48 \text{ km}$$

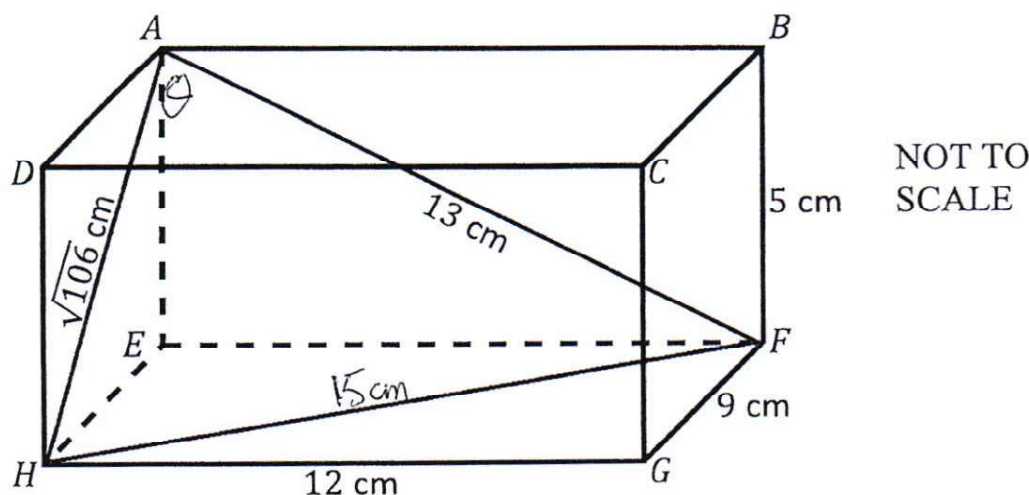
Comments: Most students used the formula, but very few with incorrect numbers.

- Many students used n as 5 (should have been 4)
- Some students wrote 13480 in km (deducted 1 mark)
I ignored the conversion from km to m.

Question 13 (3 marks)

The rectangular prism below has $BF = 5$ cm, $FG = 9$ cm, $GH = 12$ cm, $AH = \sqrt{106}$ cm and $AF = 13$ cm, as shown in the diagram below.

3



Find the size of $\angle HAF$, correct to the nearest degree.

$$\begin{aligned}
 HF^2 &= 12^2 + 9^2 \\
 &= 225 \\
 HF &= \pm \sqrt{225} \\
 &= 15 \quad (HF > 0)
 \end{aligned}
 \quad \left| \quad
 \begin{aligned}
 \cos \theta &= \frac{\sqrt{106}^2 + 13^2 - 15^2}{2 \times \sqrt{106} \times 13} \\
 &= 0.1867... \\
 \theta &= \cos^{-1}(0.1867...) \\
 &= 79^\circ
 \end{aligned}$$

Comments : Most of the students did the question correctly. However, some students used different numbers for the triangle and made the question easier.

- many students used the cosine rule correctly, however, very few swapped the values around.
- Overall well done.

Question 14 (4 marks)

- (a) A discrete random variable X has the probability distribution table shown below.

2

$X = x$	3	5	a	12
$P(x)$	0.1	m	0.2	0.4

Find the value of a and m given an expected value of 8.4.

$$0.1 + m + 0.2 + 0.4 = 1$$

$$m = 0.3$$

well done

$$3 \times 0.1 + 5 \times 0.3 + a \times 0.2 + 12 \times 0.4 = 8.4$$

$$0.2a + 6.6 = 8.4$$

$$0.2a = 1.8$$

$$a = 9$$

well done

- (b) Hence find the variance of X .

2

$$E(x^2) - \mu^2$$

$$[3^2 \times 0.1 + 5^2 \times 0.3 + 9^2 \times 0.2 + 12^2 \times 0.4] - 8.4^2$$

$$= 11.64$$

comments (b)

- majority of students did very well, except, few students input incorrect numbers, hence lost a mark.

- overall well done

Question 15 (2 marks)

Differentiate $f(x) = x^2 - 3x + 2$ using the first principles formula $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

2

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{[(x+h)^2 - 3(x+h) + 2] - [x^2 - 3x + 2]}{h} \\ & \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 3x - 3h + 2 - x^2 + 3x - 2}{h} \\ & \lim_{h \rightarrow 0} \frac{2xh + h^2 - 3h}{h} = 2x + 0 - 3 \\ & \lim_{h \rightarrow 0} 2x + h - 3 = 2x - 3 \end{aligned}$$

Comments: Very well done, except very few forgot to write h^2 while expanding.

Question 16 (4 marks)

For which values of k does $y = kx^2 + (k-2)x + 1$ have real roots? Provide your answer in simplified exact form.

4

$$\begin{aligned} & \Delta \geq 0 \\ & b^2 - 4ac \geq 0 \\ & (k-2)^2 - 4(k)(1) \geq 0 \\ & k^2 - 4k + 4 - 4k \geq 0 \\ & k^2 - 8k + 4 \geq 0 \\ & k^2 - 8k + 4 = 0 \\ & k = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(1)(4)}}{2(1)} \\ & = \frac{8 \pm \sqrt{48}}{2} = \frac{8 \pm 4\sqrt{3}}{2} = 4 \pm 2\sqrt{3} \end{aligned}$$

Comments: Many students did it well. Common mistakes include

i) $\Delta > 0$

ii) incorrect discriminant calculations

iii) choice of incorrect interval.

iv) overall, okay

$$k \leq 4 - 2\sqrt{3}, k \geq 4 + 2\sqrt{3}$$

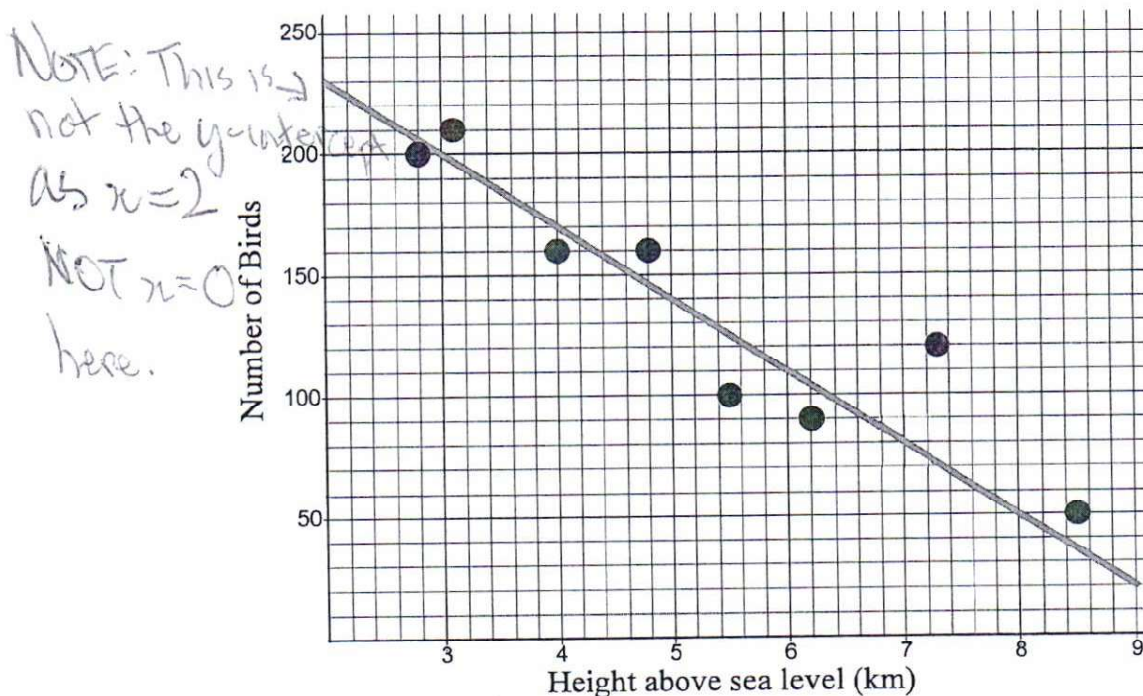
Question 17 (3 marks)

Rosanne studies the number of birds B flying above a National Park. She compares the number of birds at different heights h , in kilometres, between 2 and 9 km above sea level. She finds that the data she gathers follows a linear pattern which can be modelled by a least squares regression line with equation

$$B = -30h + c$$

where c is a constant.

Some of the data points that Rosanne collected along with the least squares regression line are shown in the scatterplot below.



Find the value of c , interpret its meaning in the given context and comment on the accuracy of this value as an estimate.

sub any point on the line. Comments: Some students wrote 230 as value of c , which is incorrect.

$$\begin{aligned} 200 &= -30(3) + c \\ c &= 290 \end{aligned}$$

Very few did not give the context or accuracy, hence lost a mark. Overall, okay.

This means at sea level there would be 290 birds. However this is extrapolation (beyond the data range) so we can't be sure of its accuracy.

Question 18 (3 marks)

Solve $5 \cos(2x + \frac{\pi}{3}) + 2 = 0$ for $0 \leq x \leq \pi$.

3

$$\cos(2x + \frac{\pi}{3}) = -\frac{2}{5}$$

$$0 \leq x \leq \pi$$

$$\text{Ref } L = \cos^{-1}(\frac{2}{5})$$

$$= 1.1071 \dots$$

$$0 \leq 2x \leq 2\pi$$

$$\frac{\pi}{3} \leq 2x + \frac{\pi}{3} \leq \frac{7\pi}{3}$$

$$\begin{array}{c|c} \text{A} & \text{B} \\ \hline \text{C} & \text{D} \end{array}$$

$$2x + \frac{\pi}{3} = \pi - 1.1071, \quad \pi + 1.1071$$

$$2x = \frac{2\pi}{3} - 1.1071, \quad \frac{2\pi}{3} + 1.1071$$

$$x = \frac{\pi}{3} - \frac{1.1071}{2}, \quad \frac{\pi}{3} + \frac{1.1071}{2}$$

$$= 0.4677, \quad 1.6267$$

Comments: Some students showed a correct working of this question. However, common mistakes include

- using degrees and radians in the same working
- using $\frac{2}{5}$ instead of $-\frac{2}{5}$ and making the question easy.

NESA Number:

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2026

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Solution

Mathematics Advanced Section II Answer Booklet 2

Booklet 2 – Attempt Questions 19 – 24 (23 marks)

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
- Write your NESA ID above

Booklet 2	Functions	Trigonometric Functions	Calculus	Exponential & Logarithmic Functions	Statistical Analysis	Financial Mathematics
	21 /2		22 /2	23 /5	19, 24 /2	20, 24 /12

Please turn over

Question 19 (2 marks)

Charlie is playing basketball. He has a 13% chance of scoring a goal from beyond the halfway line. How many shots must he take to ensure at least a 97% chance of scoring at least one goal?

2

$$\begin{aligned}
 1 - P(\text{no goal})^n &= 0.97 \\
 1 - 0.87^n &= 0.97 \\
 0.03 &= 0.87^n \\
 \log 0.03 &= n \log 0.87 \\
 n &= \frac{\log 0.03}{\log 0.87} \\
 n &= 25.18 \\
 \therefore 26 \text{ shots}
 \end{aligned}$$

1 mark - Some progress towards solution

2 marks - Correct solution

* Some students found it difficult to setup the equation at the start.

Question 20 (2 marks)

A table of future value interest factors for an annuity of \$1 is shown.

2

Rate (r) \ Period (n)	0.005	0.01	0.015	0.02	0.03	0.06
7	7.10588	7.21354	7.32300	7.43428	7.66246	8.39384
28	29.97452	32.12910	34.48148	37.05121	42.93092	68.52811
56	64.44140	74.58098	86.79754	101.55826	141.15377	418.82235
84	104.07393	130.67227	166.17264	213.86661	365.88054	2209.41674

Owais deposits \$2 000 into a savings account every three months for 7 years at an interest rate of 6% per annum, compounding quarterly.

Find the amount of money in Owais' savings account at the end of the 7 years.

$$\begin{aligned}
 r &= 0.06 \div 4 = 0.015 & n &= 7 \times 4 = 28 \\
 \$2000 \times 34.48148 &= \$68962.96
 \end{aligned}$$

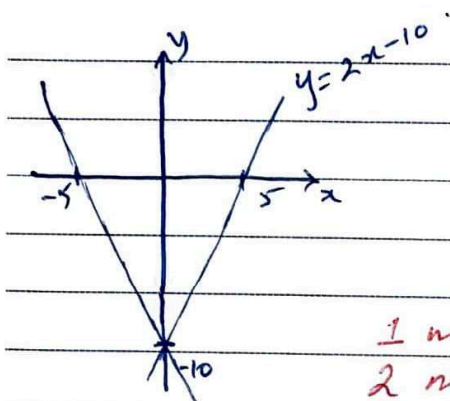
1 mark - finding the correct value of n and r.
2 marks - correct solution.

* Some students used $r = 0.06$ and $n = 7$

Question 21 (2 marks)

At the point $(3, -4)$ on the curve $y = f(x)$, the tangent has equation $y = 2x - 10$. Determine the equation of the tangent to the curve $y = f(-x)$ at the point $(-3, -4)$.

2



The tangent is reflected across the y -axis too.

$$\therefore y = -2x - 10$$

1 mark - some correct steps

2 marks - correct solution

* Many students failed to realise that the tangent is reflected in the y -axis.

Question 22 (2 marks)

Find $\int \frac{3x - 4}{3x^2 - 8x} dx$.

2

$$f(x) = 3x^2 - 8x$$

$$f'(x) = 6x - 8$$

$$\int \frac{3x - 4}{3x^2 - 8x} dx = \frac{1}{2} \int \frac{2(3x - 4)}{3x^2 - 8x} dx$$

$$= \frac{1}{2} \ln |3x^2 - 8x| + C$$

1 mark - solution had some error such as not putting the absolute value sign.

2 marks - correct solution

This question was generally done well.

Question 23 (7 marks)

- (a) Use a logarithm law to write $\ln \frac{3x-1}{x+2}$ in the form $\ln a - \ln b$, where a and b are algebraic expressions with no fractions.

1

$$\ln(3x-1) - \ln(x+2)$$

- (b) Hence, show that $y = \ln \frac{3x-1}{x+2}$ has a gradient of 1 at $x = \frac{-5 \pm \sqrt{133}}{6}$.

3

$$y = \ln(3x-1) - \ln(x+2)$$

$$y' = \frac{3}{3x-1} - \frac{1}{x+2}$$

$$1 = \frac{3}{3x-1} - \frac{1}{x+2}$$

$$= \frac{3(x+2) - 1(3x-1)}{(3x-1)(x+2)}$$

$$(3x-1)(x+2) = 3(x+2) - (3x-1)$$

$$3x^2 + 5x - 2 = 3x + 6 - 3x + 1$$

$$= 7$$

$$3x^2 + 5x - 9 = 0$$

$$x = \frac{-5 \pm \sqrt{5^2 - 4 \times 3 \times -9}}{2 \times 3}$$

$$= \frac{-5 \pm \sqrt{133}}{6}$$

Question 23 continues on the next page.

- (c) (i) Anika thought there was a fault in the answer to Question 23b. She said the answer should be $x = \frac{-5 \pm \sqrt{133}}{6}$ only. However, she is incorrect, there is no error in the answer provided in Question 23b. Anika made an error, as she was solving the question:

1

"Hence, show that $y = \ln a - \ln b$ has a gradient of 1 at $x = \frac{-5 \pm \sqrt{133}}{6}$,

Instead of the question:

"Hence, show that $y = \ln \frac{3x-1}{x+2}$ has a gradient of 1 at $x = \frac{-5 \pm \sqrt{133}}{6}$,

Provide the restriction that Anika used that lead to her incorrect answer.

$$3x-1 > 0$$

$$x+2 > 0$$

$$x > \frac{1}{3}$$

$$x > -2$$

$$x > \frac{1}{3}$$

NOTE: $x > 0$ is not the correct restriction.

- (c) (ii) Substitute $x = \frac{-5 - \sqrt{133}}{6}$ into a and b , provide answers correct to 2 decimal places, and hence explain why $x = \frac{-5 - \sqrt{133}}{6}$ is not a solution to Question 23b if you start with the equation $y = \ln a - \ln b$ but it is a solution if you start with the equation $y = \ln \frac{3x-1}{x+2}$.

2

$$a = 3\left(\frac{-5 - \sqrt{133}}{6}\right) - 1$$

$$b = \frac{-5 - \sqrt{133}}{6} + 2$$

$$= -9.27$$

$$= -0.76$$

This is not a solution using $\ln(3x-1) - \ln(x+2)$ as the arguments are both less than $\frac{1}{3}$ which is the restriction, or we can say the argument is negative but you can only take the log of a positive number. However, it is a solution using $\ln\left(\frac{3x-1}{x+2}\right)$ as the numerator and denominator are both negative, which simplifies to a positive argument.

Question 24 (10 marks)

One morning a farmer's crops were attacked by 2 000 locusts (insects). During daylight each day, including the first, the locusts population increases by 15%. Unfortunately, the farmer doesn't realise the severity of the locusts for the first two weeks.

- (a) Show that there are 14 151 locusts at the farm when the farmer realises the severity of the locust plague, to the nearest whole. 1

$$A = P(1+r)^n$$

$$= 2000(1+0.15)^{14}$$

$$= 14151.41$$

$$= 14151$$

generally done well.

- (b) The farmer does not wish to spray his farm with pesticide, so he decides to try a locust trap instead. At the start of the 15th day the farmer installs the locust trap, which traps the locust at nighttime. Find the minimum number of locusts per day that the trap must catch to reduce the number of locust from the 15th day onwards. 1

On the 15th day, there are $14151 \times 0.15 = 2122.65$ new locusts born.

$\therefore$ The trap must catch at least 2123 locusts.

generally done well. Few students made calculation errors.

- (c) Once the trap is in use, the number of locusts at the end of the n th day is given by A_n . The locust trap is guaranteed to successfully catch exactly 4 000 locusts per day. 2

- (i) Show that $A_2 = 14\,151(1.15)^2 - 4000(1.15) - 4000$.

$$A_0 = 14151$$

$$A_1 = 14151(1.15) - 4000$$

$$A_2 = [14151(1.15) - 4000](1.15) - 4000$$
$$= 14151(1.15)^2 - 4000(1.15) - 4000$$

1 mark - some progress.

2 marks - correct solution.

This question was generally done well.

Question 24 continues on the next page.

(ii) Show that $A_n = 14\,151(1.15)^n - 4000\left(\frac{1.15^n - 1}{0.15}\right)$.

3

①
$$A_3 = [14151(1.15)^2 - 4000(1.15) - 4000](1.15) - 4000$$

$$= 14151(1.15)^3 - 4000(1.15)^2 - 4000(1.15) - 4000$$

①
$$A_n = 14151(1.15)^n - 4000(1.15)^{n-1} - \dots - 4000(1.15) - 4000$$

$$= 14151(1.15)^n - 4000[1 + (1.15) + (1.15)^2 + \dots + (1.15)^{n-1}]$$

①
$$1 + (1.15) + (1.15)^2 + \dots + (1.15)^{n-1} \text{ is a G.P. } (a=1, r=1.15)$$

①
$$A_n = 14151(1.15)^n - 4000\left(\frac{1.15^n - 1}{1.15 - 1}\right)$$
 * Some students had difficulty identifying the correct series.

①
$$= 14151(1.15)^n - 4000\left(\frac{1.15^n - 1}{0.15}\right)$$

(iii) How many days since the farmer installs the trap will it take to catch all of the locusts on their farm?

3

①
$$14151(1.15)^n - 4000\left(\frac{1.15^n - 1}{0.15}\right) = 0$$

①
$$14151(1.15)^n - \frac{80,000}{3}(1.15)^n + \frac{80,000}{3} = 0$$
 * Generally done well.

$$-\frac{37547}{3}(1.15)^n = -\frac{80,000}{3}$$
 * Some students made calculation errors

$$37547(1.15)^n = 80,000$$

$$(1.15)^n = \frac{80,000}{37547}$$
 * marks were not deducted if the answer wasn't rounded to 6 days.

①
$$n \log(1.15) = \log\left(\frac{80,000}{37547}\right)$$

$$n = \frac{\log\left(\frac{80,000}{37547}\right)}{\log 1.15}$$

$$= 5.41$$

∴ 6 days

NESA Number:

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2026

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TRIAL EXAMINATION

Mathematics Advanced

Section II Answer Booklet 3

Booklet 3 – Attempt Questions 25 – 30 (22 marks)

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
- Write your NESA ID above

Booklet 3	Functions	Trigonometric Functions	Calculus	Exponential & Logarithmic Functions	Statistical Analysis	Financial Mathematics
	27 /2	25, 30 /6	28 /4		29 /8	26 /2

Please turn over

Question 25 (2 marks)

Find x , given that $\sin(x + 10)^\circ = \cos(3x + 20)^\circ$ and $0^\circ \leq x \leq 90^\circ$.

2

$$\cos(90 - (x + 10)) = \cos(3x + 20) \quad (1)$$

$$90 - x - 10 = 3x + 20$$

$$4x = 60$$

$$x = 15 \quad (1)$$

$$* \sin \theta \neq \cos(180 - \theta)$$

$$\sin \theta \neq \cos(90 + \theta)$$

Alternative:

$$(x + 10) + (3x + 20) = 90$$

$$4x + 30 = 90$$

$$x = 15$$

* Penalised for guessing and checking.

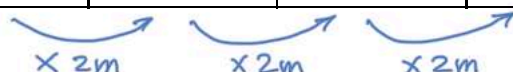
Question 26 (2 marks)

A computer program is designed to select any positive integer x . The probability distribution shown in the table below, where m is a constant, shows the probabilities of selecting the first four positive integers only.

2

x	1	2	3	4	...
$P(x)$	$3m$	$6m^2$	$12m^3$	$24m^4$	...

Find the value of m .



G.P. with $a = 3m$, $r = 2m$

For $\sum P(x) = 1$, $|r| < 1$ & $r \neq 0$.

$$S_\infty = \frac{a}{1-r}$$

$$1 = \frac{3m}{1-2m} \quad (1)$$

$$1 - 2m = 3m$$

$$5m = 1$$

$$\therefore m = \frac{1}{5} \quad (1)$$

* No marks for:

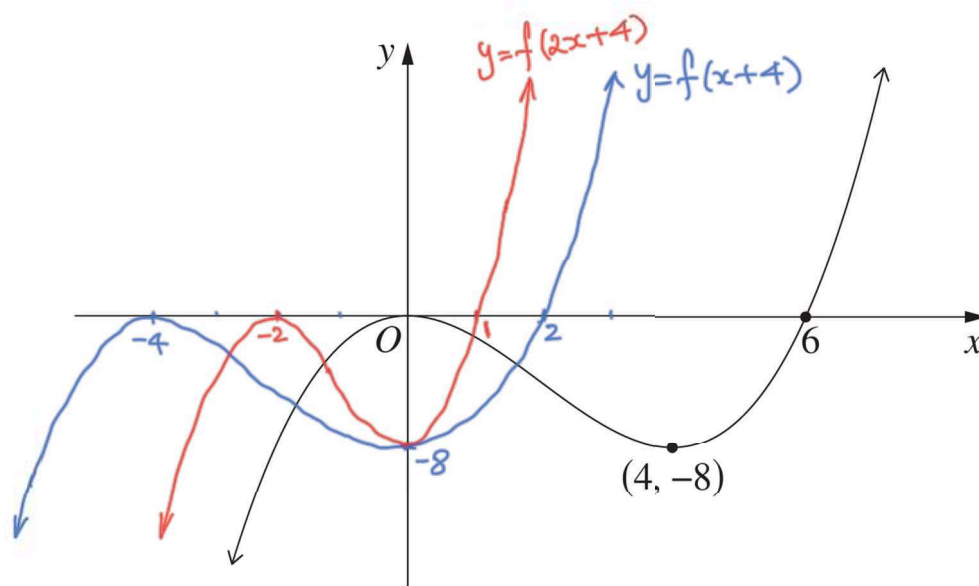
- $m = 0$
- S_∞ is not found
- Guessing and checking

Done well

Question 27 (2 marks)

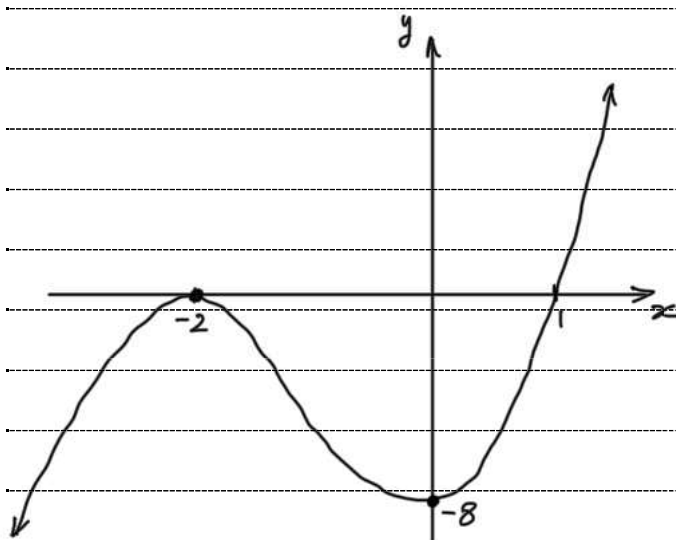
Consider the graph of $y = f(x)$ as shown.

2



NOT TO
SCALE

Sketch the graph of $y = f(2x + 4)$, labelling the x -intercepts and the coordinates of the turning points.



Note:

$f(x) \xrightarrow{\text{Left 4 units}} f(x+4) \xrightarrow{\text{Horizontal dilation by a factor of } \frac{1}{2}} f(2x+4)$

Question 28 (4 marks)

The noise level $N(t)$, in decibels during a 90 minute football match can be modelled by the function

4

$$N(t) = \frac{1}{20000} (0.25t^4 - 35t^3 + 1225t^2 + 1\,800\,000)$$

where t is the time in minutes that have been played in the match and $0 \leq t \leq 90$.

Find the loudest noise level during the football match.

$$N'(t) = \frac{1}{20000} (t^3 - 105t^2 + 2450t)$$

Let $N'(t) = 0$ for stationary points,

$$0 = \frac{1}{20000} (t^3 - 105t^2 + 2450t)$$

$$0 = t(t^2 - 105t + 2450)$$

$$0 = t(t-70)(t-35)$$

$$\therefore t = 0, 70, 35 \quad \textcircled{1}$$

$$N = 90, 90, 108.76 \quad \textcircled{1}$$

* Must check the values of the endpoints, $N(0)$ and $N(90)$.

* No marks for guessing and checking.

* Most common mistake:

Max. = 108.8

without checking $N(90)$.

$$N''(t) = \frac{1}{20000} (3t^2 - 210t + 2450)$$

$$N''(0) = \frac{1}{20000} (2450) = \frac{49}{400} > 0 \therefore \text{Min. turning point at } t=0.$$

$$N''(35) = \frac{1}{20000} (3(35)^2 - 210(35) + 2450) = -\frac{49}{800} < 0 \therefore \text{Max. turning point at } x=35.$$

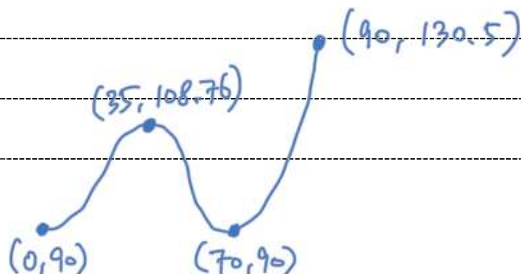
$$N''(70) = \frac{1}{20000} (3(70)^2 - 210(70) + 2450) = \frac{49}{400} > 0 \therefore \text{Min. turning point at } x=70.$$

Check end points,

$$N(0) = 90 \text{ from above}$$

$$N(90) = \frac{1}{20000} (0.25(90)^4 - 35(90)^3 + 1225(90)^2 + 1\,800\,000) = 130.5 \quad \textcircled{1}$$

$\therefore$ The loudest noise level is 130.5 decibels. $\textcircled{1}$



Can be skipped & just compare N values.

Question 29 (8 marks)

Elena, the owner of Elena's Escape Rooms, records the time taken in minutes for each group to complete her escape room. She recognises this data as a continuous random variable, X , which can be modelled by the probability density function

$$f(x) = \begin{cases} -\frac{1}{4500}(x-10)(x-40) & 10 \leq x \leq 40 \\ 0 & \text{elsewhere} \end{cases}$$

Done well

- (a) Find the mode time taken to complete the escape room at Elena's Escape Rooms.

$$f(x) = -\frac{1}{4500}(x^2 - 50x + 400)$$

$$f'(x) = -\frac{1}{4500}(2x - 50)$$

Let $f'(x) = 0$ for stationary point,

$$0 = -\frac{1}{4500}(2x - 50)$$

$$x = 25$$

Since $a = -\frac{1}{4500} < 0$ $f(x)$ is concave down.

$\therefore f(x)$ is max. at $x = 25$.

$\therefore \text{Mode} = 25$.

Alternative

Sub $f(x) = 0$ for x -intercepts,
 $0 = -\frac{1}{4500}(x-10)(x-40)$

$$x = 10, 40$$

$f(x)$ is concave down parabola
 Since it is quadratic & $a < 0$.

Max. $f(x)$ occurs at $x = \frac{10+40}{2}$
 $= 25$
 $\therefore \text{Mode} = 25$

* Mode is the x value

when $f(x)$ is maximum.

Mode $\neq f(x)$.

* No marks if no steps shown.

- (b) (i) Show that the cumulative density function for the time taken to complete the escape room at Elena's Escape Rooms is given by

2

Done well

$$F(x) = -\frac{1}{4500}\left(\frac{x^3}{3} - 25x^2 + 400x\right) + \frac{11}{27}$$

$$F(x) = -\frac{1}{4500} \int_{10}^x (x^2 - 50x + 400) dx \quad \textcircled{1}$$

$$= -\frac{1}{4500} \left[\frac{x^3}{3} - 25x^2 + 400x \right]_{10}^x$$

$$= -\frac{1}{4500} \left(\frac{x^3}{3} - 25x^2 + 400x - \left(\frac{10^3}{3} - 25(10)^2 + 400(10) \right) \right)$$

$$= -\frac{1}{4500} \left(\frac{x^3}{3} - 25x^2 + 400x - \frac{5500}{3} \right)$$

$$= -\frac{1}{4500} \left(\frac{x^3}{3} - 25x^2 + 400x \right) + \frac{11}{27}$$

Questions 29 continues on the next page.

Done well

- (b) (ii) Elena used the cumulative density function found above and an online calculator to find the 25th and 75th percentiles for her escape room. Two screenshots of the calculations made by the online calculator are shown below.

1

Screenshot 1:

$\text{solve } 0.25 = -\frac{1}{4500} \left(\frac{x^3}{3} - 25x^2 + 400x \right) + \frac{11}{27}$

Steps Graph Examples

$0.25 = -\frac{1}{4500} \left(\frac{x^3}{3} - 25x^2 + 400x \right) + \frac{11}{27}$

Solution

$x \approx 2.01866..., x \approx 19.79055..., x \approx 53.19077...$

Since $10 \leq x \leq 40$

Screenshot 2:

$\text{solve } 0.75 = -\frac{1}{4500} \left(\frac{x^3}{3} - 25x^2 + 400x \right) + \frac{11}{27}$

Steps Graph Examples

$0.75 = -\frac{1}{4500} \left(\frac{x^3}{3} - 25x^2 + 400x \right) + \frac{11}{27}$

Solution

$x \approx -3.19077..., x \approx 30.20944..., x \approx 47.98133...$

Since $10 \leq x \leq 40$

Using the screenshots above, calculate the interquartile range for the time taken to complete Elena's Escape Room.

$$IQR = Q_3 - Q_1$$

$$= 30.20944 - 19.79055 \quad (\text{Since } 10 \leq x \leq 40)$$

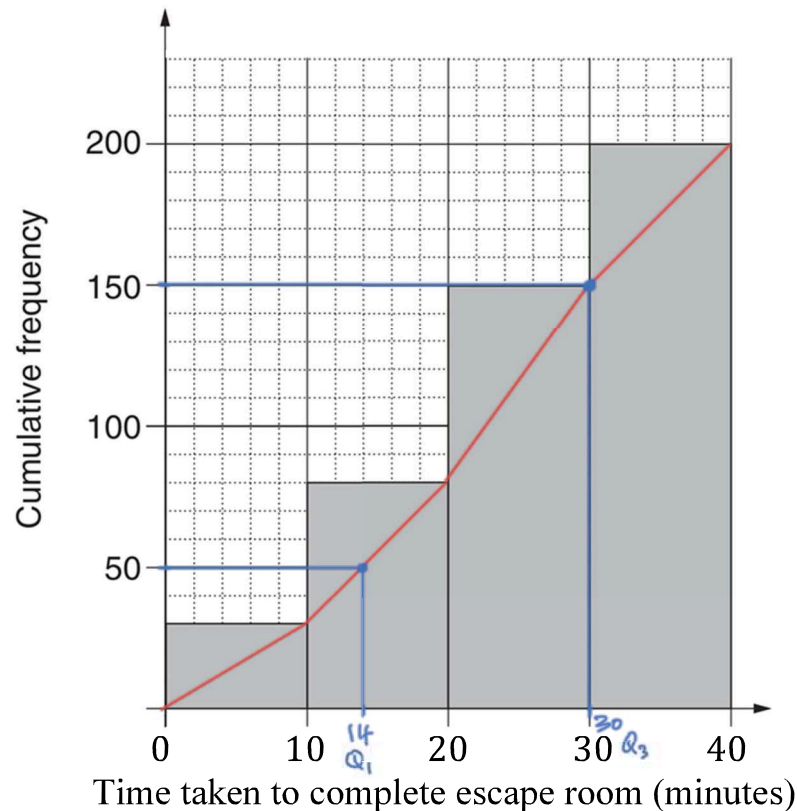
$$= 10.41889$$

$$= 10.42 \text{ min (2 d.p.)}$$

Questions 29 continues on the next page.

- (c) (i) A different escape room company, Escape From Axel, also gathers data on the time its customers take to complete their escape room. Axel displays the data in the cumulative frequency histogram below.

Poorly done



Calculate the interquartile range for the time taken to complete the escape room at Escape From Axel.

$$IQR = Q_3 - Q_1$$

$$= 30 - 14$$

$$= 16 \text{ minutes}$$

* 1 mark for either Q_1 or Q_3

* No marks if both Q_1 and Q_3 are incorrect.

* Time is a continuous variable. Use the ogive for Q_1 and Q_3 .

Done well

- (c) (ii) Find the modal time group for the time taken to complete the escape room at Escape From Axel.

20 - 30 minutes

* Most common mistake: modal group = 30.

Done well

Question 30 (4 marks)

Identify the period, amplitude, phase shift and vertical shift of the graph formed by the equation

4

$$y = 12\cos\left(\frac{\pi}{3}(x - 2)\right) + 4.$$

$$\text{Amplitude} = 12$$

$$\text{Vertical shift} = 4 \text{ units up}$$

$$\text{Period} = \frac{2\pi}{\frac{\pi}{3}}$$

$$= 6$$

$$\text{Phase shift} = 2 \text{ units to the right.}$$

* Penalised if 'phase shift' is not stated.

"Horizontal shift by 2 units to the right" is not sufficient, as

it does not demonstrate

whether you understand the

meaning of phase shift or

simply describing every translation.

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Mathematics Advanced

Section II Answer Booklet 4

Booklet 4 – Attempt Questions 31 – 35 (22 marks)

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
- Write your NESA ID above

Booklet 4	Functions	Trigonometric Functions	Calculus	Exponential & Logarithmic Functions	Statistical Analysis	Financial Mathematics
		32, 35 /10	33 /4	31 /4		34 /4

Please turn over

Question 31 (4 marks)

Given $f(x) = 16^x$ and $g(x) = \frac{1}{2}x + \frac{3}{4}$ show that the derivative of $y = f(g(x))$ is $\ln 2 (2^{2x+4})$.

4

$$f(g(x)) = 16^{\frac{1}{2}x + \frac{3}{4}}$$

$$\begin{aligned} f'(g(x)) &= \frac{1}{2} \times 16^{\frac{1}{2}x + \frac{3}{4}} \times \ln 16 \\ &= \frac{1}{2} \times (2^4)^{\frac{1}{2}x + \frac{3}{4}} \times \ln 2^4 \\ &= \frac{1}{2} \times 2^{2x+3} \times 4 \ln 2 \\ &= 2^{2x+3} \times 2 \ln 2 \\ &= 2^{2x+3} \times 2 \times \ln 2 \\ &= 2^{2x+3+1} \times \ln 2 \\ &= \ln 2 (2^{2x+4}) \end{aligned}$$

ALTERNATE METHOD

$$\begin{aligned} f(g(x)) &= 16^{\frac{1}{2}x + \frac{3}{4}} \\ &= (2^4)^{\frac{1}{2}x + \frac{3}{4}} \\ &= 2^{2x+3} \end{aligned}$$

$$\begin{aligned} f'(g(x)) &= 2 \times 2^{2x+3} \times \ln 2 \\ &= 2^1 \times 2^{2x+3} \times \ln 2 \\ &= 2^{2x+3+1} \ln 2 \\ &= \ln 2 (2^{2x+4}) \end{aligned}$$

Question 32 (3 marks)

Find $\int \frac{\tan^4 x}{1 - \sec^2 x} dx$.

3

Writing this led to fewer errors.
Many made the mistake of:
 $\tan^2 x = 1 - \sec^2 x$.

$$= \int \frac{\tan^4 x}{1 - \sec^2 x} dx$$

$$= \int -\tan^2 x dx$$

$$= \int 1 - \sec^2 x dx$$

$$= x - \tan x + C$$

$$\begin{aligned} 1 + \tan^2 x &= \sec^2 x \\ \tan^2 x &= \sec^2 x - 1 \\ -\tan^2 x &= 1 - \sec^2 x \end{aligned}$$

NOTE: Many students made errors with negatives. Some made two errors, which cancelled each other out and lead to the correct answer. However, two errors were made so marks were lost despite the correct answer.

ALTERNATE METHODS:

$$\begin{aligned} &\int -\tan^2 x dx \\ &= -\int \sec^2 x - 1 dx \\ &= -(\tan x - x) + C \\ &= x - \tan x + C \end{aligned}$$

$$\begin{aligned} &\int \frac{(\tan^2 x)^2}{1 - \sec^2 x} dx \\ &= \int \frac{(-\tan^2 x)^2}{1 - \sec^2 x} dx \\ &= \int \frac{(1 - \sec^2 x)^2}{1 - \sec^2 x} dx \\ &= \int 1 - \sec^2 x dx \\ &= x - \tan x + C \end{aligned}$$

$$\begin{aligned} &\int \frac{(\tan^2 x)^2}{1 - \sec^2 x} dx \\ &= \int \frac{(\sec^2 x - 1)^2}{1 - \sec^2 x} dx \\ &= -\int \frac{(\sec^2 x - 1)^2}{\sec^2 x - 1} dx \\ &= -\int \sec^2 x - 1 dx \\ &= -(\tan x - x) + C \\ &= x - \tan x + C \end{aligned}$$

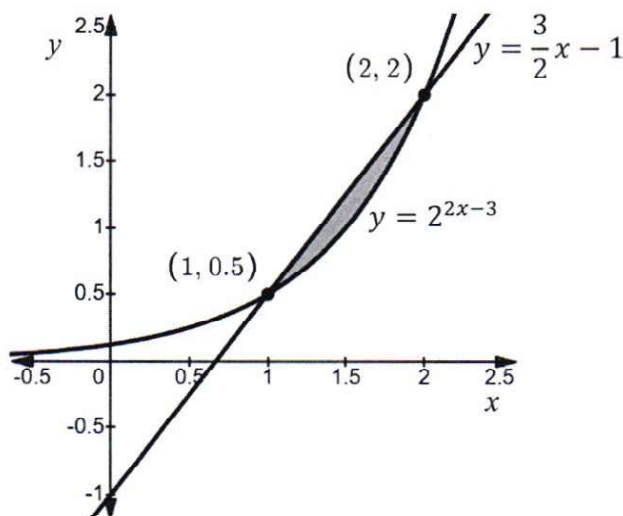
NOTE: $\tan^4 x \neq 1 - \sec^4 x$.

These formulae are for powers of 2 not 4.

Question 33 (4 marks)

The curve $y = 2^{2x-3}$ meets the line $y = \frac{3}{2}x - 1$ at $(1, 0.5)$ and $(2, 2)$, as shown in the diagram.

4



Show that the shaded region has an area of $\frac{5\ln 2 - 3}{4\ln 2}$ units².

$$\int_1^2 \left(\frac{3}{2}x - 1 \right) dx - \int_1^2 2^{2x-3} dx$$

$$\left[\frac{3}{4}x^2 - x \right]_1^2 - \frac{1}{2} \int_1^2 2 \times 2^{2x-3} dx$$

$$\left(\frac{3}{4}(2)^2 - (2) \right) - \left(\frac{3}{4}(1)^2 - (1) \right) - \frac{1}{2} \left[\frac{2^{2x-3}}{\ln 2} \right]_1^2$$

$$\frac{5}{4} - \frac{1}{2} \left(\frac{2^{2(2)-3}}{\ln 2} - \frac{2^{2(1)-3}}{\ln 2} \right)$$

$$\frac{5}{4} - \frac{1}{2} \left(\frac{2}{\ln 2} - \frac{2^{-1}}{\ln 2} \right)$$

$$\frac{5}{4} - \frac{2}{2\ln 2} + \frac{1}{4\ln 2}$$

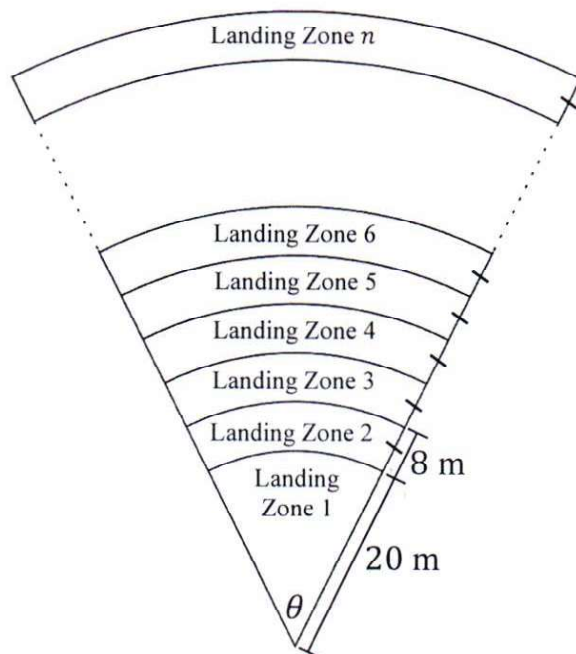
$$\frac{5\ln 2}{4\ln 2} - \frac{4}{4\ln 2} + \frac{1}{4\ln 2} = \frac{5\ln 2 - 3}{4\ln 2}$$

NOTE: Less errors were made when the integrals were kept separate rather than subtracting the functions first then integrating.

Question 34 (4 marks)

Mrs Palmer is setting up a javelin landing zone. The landing zone takes the form of a sector with angle θ in radians. The first arc is 20 metres from the vertex of the sector and consecutive arcs are 8 metres apart. This is shown in the diagram below.

4



The perimeter and internal arcs separating each landing zone must be marked on the grass with black paint, as shown in the diagram. Show that the total length of the lines to be marked with paint can be given by $L_n = 4(n^2\theta + 4n\theta + 4n + 6)$, where n is the number of landing zones.

$$\begin{aligned} \text{Straight lines} &\rightarrow = [20 + (n-1)8] \times 2 \\ &= [20 + 8n - 8] \times 2 \\ &= [12 + 8n] \times 2 \\ &= 24 + 16n \end{aligned}$$

$$\begin{aligned} \text{Arcs} &\rightarrow 20\theta + 28\theta + 36\theta + \dots + [20 + (n-1)8]\theta \\ &= \theta (20 + 28 + 36 + \dots + (20 + (n-1)8)) \end{aligned}$$

$$\begin{aligned} &\text{sum of AP, } a=20, d=8 \\ &= \theta \left(\frac{n}{2} (2 \times 20 + (n-1)8) \right) \end{aligned}$$

$$= \theta \left(\frac{n}{2} (40 + 8n - 8) \right)$$

$$= \theta \left(\frac{n}{2} (32 + 8n) \right)$$

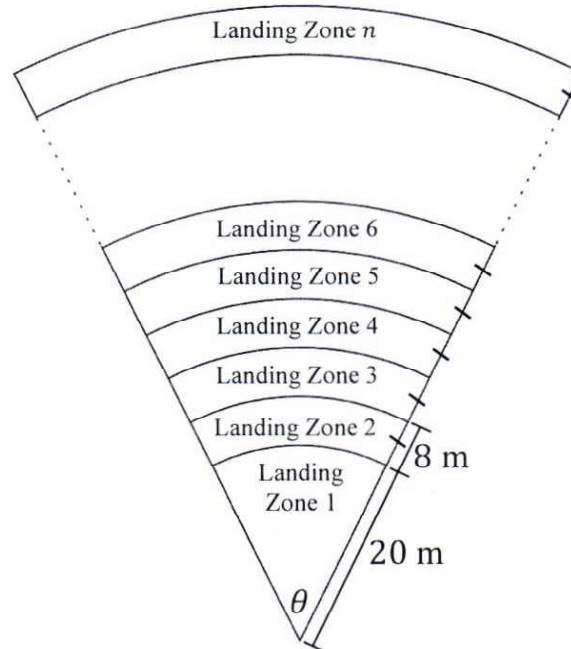
$$= 16n\theta + 4n^2\theta$$

$$\begin{aligned} \text{Total} &= 24 + 16n + 16n\theta + 4n^2\theta \\ &= 4(n^2\theta + 4n\theta + 4n + 6) \end{aligned}$$

Question 34 (4 marks)

Mrs Palmer is setting up a javelin landing zone. The landing zone takes the form of a sector with angle θ in radians. The first arc is 20 metres from the vertex of the sector and consecutive arcs are 8 metres apart. This is shown in the diagram below.

4



The perimeter and internal arcs separating each landing zone must be marked on the grass with black paint, as shown in the diagram. Show that the total length of the lines to be marked with paint can be given by $L_n = 4(n^2\theta + 4n\theta + 4n + 6)$, where n is the number of landing zones.

ALTERNATE METHOD

$$P_1 = 2 \times 20 + 20\theta$$

$$= 40 + 20\theta$$

$$P_2 = 40 + 20\theta + 8 \times 2 + (20 + 8)\theta$$

$$= 40 + 16 + 20\theta + 28\theta$$

$$P_3 = 40 + 16 + 20\theta + 28\theta + 8 \times 2 + (20 + 8 \times 2)\theta$$

$$= 40 + 16 \times 2 + 20\theta + 28\theta + 36\theta$$

$$P_n = 40 + 16(n-1) + \theta(20 + 28 + 36 + \dots + [20 + (n-1)8])$$

$AP, a=20, l=20+(n-1)8$

$$= 40 + 16n - 16 + \theta \left(\frac{n}{2} (20 + 20 + (n-1)8) \right)$$

$$= 24 + 16n + \theta \left(\frac{n}{2} (40 + 8n - 8) \right)$$

$$= 24 + 16n + \theta \left(\frac{n}{2} (32 + 8n) \right)$$

$$= 24 + 16n + 16n\theta + 4n^2\theta$$

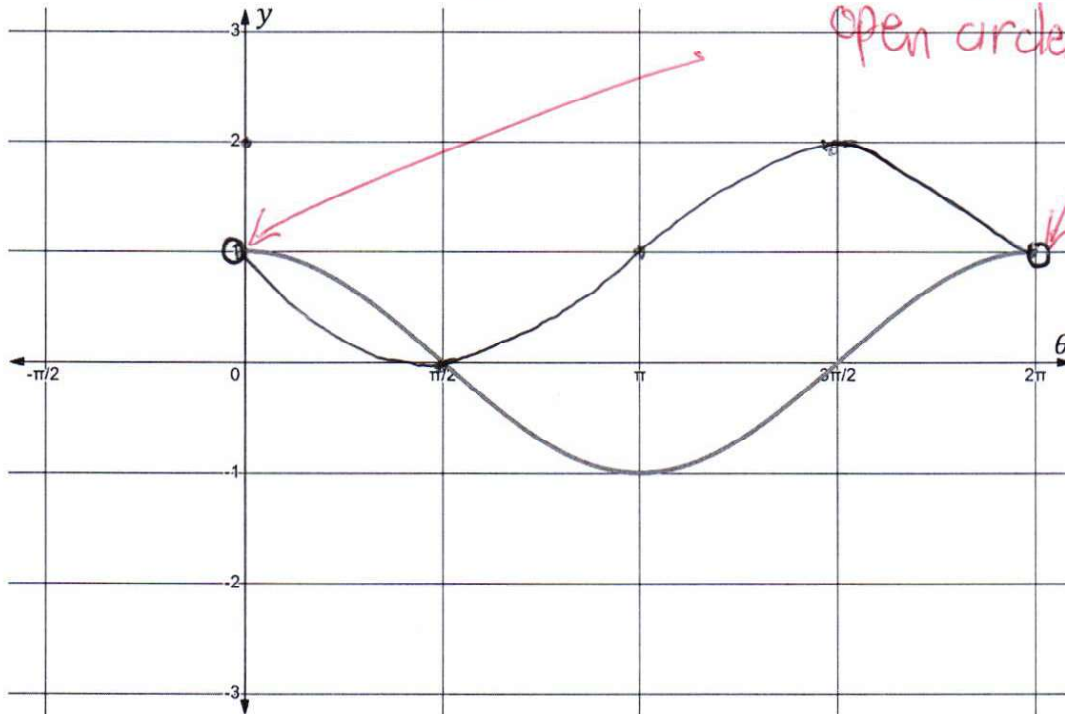
$$= 4(n^2\theta + 4n\theta + 4n + 6)$$

Question 35 (7 marks)

- (a) The graph of $y = \cos \theta$ for $0 \leq \theta \leq 2\pi$ is shown below. On the same Cartesian Plane, sketch the graph of $y = -\sin \theta + 1$ for $0 < \theta < 2\pi$.

3

NOTE: Most students forgot the open circles on the ends.



- (b) Hence or otherwise, find any point(s) of intersection of $y = \cos \theta$ and $y = -\sin \theta + 1$ for $0 < \theta < 2\pi$.

2

$\pi/2$

CFE's were given if open circles were missed on the first part.

Question 35 continues on the next page

(c) Hence or otherwise, solve $\sin^2 \theta + \sin \theta \cos \theta - \sin \theta = 0$ for $0 < \theta < 2\pi$.

2

$$\sin \theta (\sin \theta + \cos \theta - 1) = 0$$

$$\sin \theta = 0$$

$$\sin \theta + \cos \theta - 1 = 0$$

$$\theta = \pi$$

$$\cos \theta = -\sin \theta + 1$$

$$\theta = \frac{\pi}{2} \text{ (from part ii)}$$